

Deliberation and Learning: Evidence From Deliberative Polls*

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Abstract

We study factual learning among 8,450 participants in 622 discussion groups across 27 Deliberative Polls held between 1994 and 2021. The polls address elections, energy, health, crime, climate, and other public issues in Europe, Australia, China, and the United States. Across polls, absolute gains averaged 14.6 percentage points and relative gains 40.2%. An item model assuming no forgetting yields estimated learning of 19.4 percentage points. In two polls with control groups, attendees gained an average of 11.2 percentage points more than controls. In five studies, we compare knowledge before arrival, on arrival, and after deliberation. Substantial gains occur before participants arrive.

1 Introduction

Among the great virtues commonly ascribed to deliberation is learning. The participants learn about the topics under discussion and, to a lesser degree, incidentally, about related ones. They learn not only relevant facts but the arguments for and against particular alternatives and what other people unlike themselves, their circumstances, and their ways of thinking are like. This is helpful, both at the individual level, for discerning what policies most closely accord with one's values and interests and voting accordingly, and at the aggregate level, for forming "authentic" majorities—majorities that "know what they are doing" (Luskin 2003). How much citizens know shapes what they prefer and how they vote (Converse 1964; Bartels 1996; Althaus 1998; Delli Carpini and Keeter 1996), even on foreign policy (Aldrich et al. 1989; Foyle 2004), though information shortcuts can partly stand in for knowledge (Popkin 1991; Lupia 1994).

The expectation of learning follows directly from the nature of "deliberation," conventionally construed as discussion, but not just any discussion (Fishkin and Luskin 2005). While definitions vary, deliberation is generally viewed as normatively elevated discussion—in which, among other things, the participants clarify and refine their preferences by open-mindedly weighing competing arguments. As part of that weighing, they absorb factual information and sift out misinformation.

Or so the theory runs. But how much do participants actually learn, is it the deliberation that teaches them, and who learns the most? We examine learning of relevant facts. This is not the only sort of learning involved, but it is the easiest to measure, and it is both important in itself and positively correlated with the rest.

We compare participants' initial and later knowledge, using one sample for the main learning estimates and discussion-group models. Interview timing varies across studies. Where measurements before arrival, on arrival, and after deliberation are available, we separately measure changes before

*Data and replication materials: <https://github.com/soodoku/dp-learning>.

arrival and during the event among the same participants. We also compare pre-arrival knowledge by invitation and attendance to assess selection into the observed participant sample. In America in One Room, we compare attendees with nonattendees in the recruitment sample and with a separate control group not invited to deliberate. In the climate study, the corresponding participant group consists of people classified as completing the deliberations; other invitees may have participated in some sessions. Before-and-after knowledge scores are available for these participants and controls; the remaining recruitment or invitation sample contributes only initial scores. Earlier studies document learning in Deliberative Polls and other forums (Luskin et al. 2002; Barabas 2004; Sturgis et al. 2005; Esterling et al. 2011; Theuwis et al. 2025). We report gains in both absolute and relative terms, adjust for guessing, and examine whether participants learn more alongside more knowledgeable groupmates.

2 Deliberative Polling and Learning

Deliberative Polling (Fishkin 1991, 1995; Luskin et al. 2002) aims to capture fuller-information, more thoughtful opinions than those expressed in ordinary polls. In very brief, a Deliberative Poll interviews and assembles, as many as are willing of, a random sample to deliberate some policy or electoral choice, in alternating small group discussions and plenary question-and-answer sessions with panels of competing experts, and then re-measures the sample’s opinions at the end. These latter are frequently different, both at the individual level and in the aggregate, from what they were before deliberating. The absolute net change is statistically significant nearly three-quarters of the time and averages nearly .1 on a 0–1 scale (Luskin et al. 2007).

This design, admittedly a particular embodiment of deliberation, affords additional opportunities and incentives to learn. The participants are provided with carefully balanced briefing documents, rehearsing the arguments each side would make. There is usually a section of basic background information, and the arguments embed each side’s pet facts—those most congenial to its arguments. In addition, the participants know from the moment they are first recruited that they will be attending the deliberation and that the deliberation (typically) will be an “Event,” capital “e”—away from home, usually televised or taped for later broadcast, often attended by high-profile policy experts or policy-makers in the role of observers or panelists. Thus, motivated by a desire to “do well,” many participants use the weeks between their recruitment and the event itself to prepare—not only by reading the briefing materials but by paying heightened attention to stories about the topic in the media, by talking more with friends, family, and coworkers about it, and even, in some cases, by researching it in the library or online.

So if knowledge, like its close relative “sophistication,” depends on the confluence of cognitive ability, motivation, and exposure to relevant information (Luskin 1990, 2002), the deliberation in Deliberative Polling stands to increase two of those three. There is not much to be done, in adulthood, about ability—and in fact we should want the participants to be representative in this, as in all other respects. But the treatment can and does affect motivation and exposure. Judging from past polls, that is enough to produce considerable, sometimes dramatic learning (Luskin et al. 2002; List et al. 2013).

The online design is intended to be as similar as possible to the face-to-face one, given the difference in mode. The earlier online polls used voice discussion to capture as much as possible of deliberation’s social elements.¹ In these earlier polls, participants still “met” in small groups, led by trained

¹More recent online Deliberative Polls use live video discussion. The 2021 America in One Room climate poll

moderators, and questioned panels of policy experts and policy makers. They still receive briefing materials. They are still a representative sample—sometimes randomly sampled offline, with “nonusers” receiving free access, including free equipment, to get them online, sometimes drawn from online panels and matched to approximate a random sample. On the other hand, deliberating by voice, even by video, online can never capture the full richness and complexity of in-person interactions; the discussions are interrupted, spread over a period of weeks, with much greater possibility of real-world intrusion, and the assignment to small groups is perforce constrained by possible meeting times. All these differences may act to weaken the online treatment’s effects (Luskin et al. 2006; Iyengar et al. 2005).

3 Theory and hypotheses

Learning requires exposure to information and the ability and motivation to absorb it. These mechanisms suggest when participants should learn as well as who should learn most.

When learning occurs. We expect some learning before participants arrive. The briefing materials provide information, and the prospect of discussing the issues gives participants a reason to absorb it. Invitation can also prompt them to seek information elsewhere. Thus preparation should raise knowledge between recruitment and arrival, even among some invitees who ultimately do not attend.

We expect further learning during deliberation. Discussion gives participants opportunities to ask questions, encounter facts they missed, and correct misunderstandings. Expert answers and explanations from other participants can make difficult material easier to grasp. Preparation may help participants learn from these exchanges by giving them the background needed to understand them. But it also leaves less room for further gains on a bounded knowledge test. We therefore expect gains both before arrival and during the event, without predicting which will be larger. Smaller gains during the event need not mean that discussion offers little: they may partly reflect what participants have already learned.

After the event, the question becomes how much of the gain lasts. If participants stop encountering or using the information, we expect some of it to be forgotten. Continued attention and discussion may instead sustain knowledge or add to it. Later interviews therefore address the durability of the gains. Changes since exit reflect both forgetting and subsequent learning, rather than forgetting alone.

The same mechanisms motivate our expectations about individual and group differences.

Initial knowledge. The information-rich tend to get information-richer. The more one knows going in, the more one tends to learn and thus know going out. Absent ceiling effects, this predicts a positive association with gain; bounded scores can instead produce a negative association. Prior knowledge on a particular topic improves people’s ability to comprehend new information, enabling them to extract the central elements of a message and draw appropriate inferences efficiently (Eckhardt et al. 1991; Recht and Leslie 1988). Prior knowledge also enhances people’s ability to store new information on that topic and retrieve it later (Cooke et al. 1993; Fiske et al. 1990; Hambrick 2003; McGraw and Pinney 1990; Recht and Leslie 1988; Schneider et al. 1993). So the less political information individuals have stored in memory, the more difficult it is for them to acquire new information.

used the Stanford Online Deliberation Platform, with an automated moderator managing speaking turns and the discussion agenda (Fishkin et al. 2025, 436–37).

Education. Education, qua years of schooling, is distinct from, and only moderately correlated with, political knowledge. Controlling for cognitive ability, with which it is more strongly correlated, its effect on knowledge appears negligible (Luskin 1990). But absent a separate measure of cognitive ability, it can roughly proxy the latter. Naturally, we expect it to have a positive effect. There may, however, be some mutual substitution with initial knowledge. People who know enough already may not need to be as bright to learn a lot, and people who are bright enough may not need to know as much to do so.

Gender and age. These demographic variables exhibit real-world political knowledge gaps, with men knowing more than women and older people more than younger ones (Delli Carpini and Keeter 1996). These differences can reflect prior opportunities to acquire information. Conditional on initial knowledge, their association with learning during deliberation need not have the same direction. We include both as participant characteristics without imposing a directional hypothesis.

Gender composition. Whether from greater comfort, more widely shared cognitive or communication style, or lower stereotype threat (Steele 1997), women may learn more in groups containing more women. We examine this possibility by interacting gender with the proportion of female groupmates.

Reading the briefing materials. Reading provides exposure to the facts and competing arguments discussed at the event. We therefore expect participants who read more of the materials to learn more, although motivation may influence both reading and learning.

Group knowledge. More knowledgeable groupmates can supply facts, correct errors, and explain unfamiliar arguments. We therefore expect their initial knowledge to be positively associated with a participant’s gain, conditional on that participant’s own initial knowledge.

Group size and mode. Larger groups may offer more perspectives but give each participant less time to speak (Latané et al. 1979), leaving the expected association with learning ambiguous. Online deliberation may offer thinner exposure to other participants and weaker opportunities to learn than in-person discussion (Luskin et al. 2006).

Attitudes. Strong policy preferences may motivate attention to relevant information or make conflicting information harder to accept. Disagreement within a group may expose participants to unfamiliar arguments, but it can also impede communication. We examine attitude extremity and group disagreement without imposing a directional prediction.

4 Research design and model

4.1 Interview timing and the quantities of interest

A Deliberative Poll begins by recruiting citizens, usually through a survey conducted before they receive briefing materials. Those who participate can prepare before arriving, then deliberate in small groups and question experts. Knowledge may be measured at four stages: before arrival (t_0), on arrival before discussion (t_1), at the end of deliberation (t_2), and at a later follow-up (t_3). For online polls, arrival means the start of the discussion sessions. The labels refer to these stages in every poll; a missing arrival questionnaire does not turn a pre-arrival interview into t_1 . A questionnaire administered partway through deliberation, as in New Haven, is an interim measurement.

These stages distinguish three quantities. The change $K_{t_1} - K_{t_0}$ measures gains before arrival,

when participants may read the briefing materials or prepare in other ways. The change $K_{t_2} - K_{t_1}$ measures gains during the event. The change $K_{t_2} - K_{t_0}$ includes both periods. Follow-up measurements at t_3 address how much knowledge remains later. Each is a change among the same people answering the same questions. None by itself identifies a causal effect: repeated testing and learning outside the study can also change scores.

Our main pooled estimate is the average within-poll change among attendees, giving each poll equal weight. It compares pre-arrival and exit (t_0 to t_2), covering both preparation and the event. The separate three-wave comparisons distinguish gains before arrival from gains during deliberation. Later follow-ups address how much knowledge persists and are analyzed separately.

4.2 Invitation, attendance, and causal effects

The two studies with controls, America in One Room and its climate successor, allow us to subtract controls' pre-arrival-to-exit change from attendees' change. This difference removes changes common to both groups. It identifies the effect of participation on attendees only if, without participation, attendees and controls would have had the same average change. Initial differences in knowledge alone do not violate that assumption, but differences in how the two groups would otherwise learn could do so.

An intention-to-treat (ITT) effect concerns invitation, including any preparation it induces before the event. With randomized invitations and follow-up outcomes for all invitees and controls, it could be estimated without assuming that nonattenders learned nothing. Our participant-control comparisons do not estimate that effect: follow-up knowledge scores are unavailable for the remaining recruitment or invitation sample, and the 2019 recruitment indicator does not establish individual invitation status. Assigning nonattenders zero gains would be an assumption about missing outcomes. Assuming instead that invitation had no effect on their learning would still allow the gains they would have made without it.

Under random assignment, an invitation that increases attendance, no interference between participants, and the exclusion and monotonicity assumptions, dividing the ITT effect by the invitation-induced change in attendance identifies a local average treatment effect (LATE) for people whose attendance is induced by invitation (Angrist et al. 1996). The exclusion assumption requires invitation to affect knowledge only through attendance. That is questionable here. The climate study sent electronic briefing materials to registered delegates before deliberation, so recipients could read them without eventually attending (NORC at the University of Chicago 2021, 5–6). An invitation could also prompt independent preparation. Such learning belongs in the effect of invitation but prevents interpreting the usual instrumental-variable ratio as the effect of attendance alone.

4.3 Model

Let $K_{igd,t}$ denote the proportion of knowledge questions answered correctly by participant i in group g and poll d at stage t . Our outcome is the gain from pre-arrival to exit, $G_{igd} = K_{igd,t_2} - K_{igd,t_0}$. We fit a hierarchical linear model (Raudenbush and Bryk 2002):

$$G_{igd} = \beta_0 + \beta_1 K_{igd,t_0} + \mathbf{z}'_{igd} \beta_z + \alpha_{gd} + \gamma_d + \varepsilon_{igd},$$

where $\mathbf{z}_{igd}$ contains groupmates' mean initial knowledge, group size, and mode. The second specification adds education, age, gender, the proportion of female groupmates, and the education-by-initial-knowledge and gender-by-group-composition interactions. The third and fourth specifications separately extend the demographic model: the third adds attitude extremity and group

disagreement, and the fourth adds reported briefing reading. The β coefficients are fixed regression parameters. For fitting, we assume

$$\alpha_{gd} \sim N(0, \sigma_g^2), \quad \gamma_d \sim N(0, \sigma_p^2), \quad \varepsilon_{igd} \sim N(0, \sigma_e^2),$$

with these components mutually independent conditional on the regressors and independent across distinct groups, polls, and individuals, respectively. The group and poll random intercepts allow outcomes to covary within discussion groups and across groups in the same poll. Their variances, σ_g^2 and σ_p^2 , measure residual variation at those levels; σ_e^2 measures individual residual variation. We estimate the coefficients and variance components by maximum likelihood (Bates et al. 2015) and obtain confidence intervals from the hierarchical bootstrap.

Adding K_{igd,t_0} to both sides gives the equivalent equation for later knowledge:

$$K_{igd,t_2} = \beta_0 + (1 + \beta_1)K_{igd,t_0} + \mathbf{z}'_{igd}\beta_z + \alpha_{gd} + \gamma_d + \varepsilon_{igd}.$$

Thus using gain instead of the later score shifts the coefficient on initial knowledge down by one and leaves the other coefficients and fitted residuals unchanged. We report gain because it directly expresses the change the paper seeks to explain. Conditioning on initial knowledge compares that change among participants with the same measured starting knowledge.

An unadjusted change-score model instead constrains the baseline coefficient to zero in the gain equation. These specifications can disagree when groups differ at baseline (Allison 1990; van Breukelen 2006). Score ceilings, guessing, and measurement error complicate the relationship between observed gains and learning of the underlying facts (Luskin et al. 2016). We therefore treat the initial-knowledge coefficient as an association, not an estimate of how prior knowledge causes learning.

The comparisons follow these distinctions. We first describe who attends and how attendees differ from the comparison groups before the event. We then report absolute and relative knowledge gains, compare attendees with controls, distinguish gains before arrival from gains during the event, and examine how much knowledge persists at later follow-ups. Finally, we use the model to assess the hypotheses about participants' initial knowledge, education, and other characteristics; the knowledge and composition of their discussion groups; and their attitudes and reading of the briefing materials.

5 Data and measures

The main sample comprises 27 polls and 8,450 attendees with initial and later knowledge scores and recorded discussion-group membership. It supports both the learning estimates and the analysis of discussion groups. We use the Deliberative Poll dataset (Sood 2026), linking each participant's answers across interviews. Each person contributes one paired observation within a poll; overlapping datasets for the same study do not supply additional participants. Appendix D gives the questions and scoring rules.

The two datasets for the 2004 online Primaries describe the same experiment and include overlapping respondents. We include that study once, using attendees with at least one documented discussion session and observed initial and exit questionnaires. This gives 239 paired attendees, of whom 238 have known groups and enter the main models. The 217-person historical sample is an overlapping subset of that study.

Comparisons by interview timing require documented attendance, evidence that respondents answered each questionnaire, and the same knowledge questions and scoring denominator at both interviews. The main analysis requires observed questionnaires at every stage through immediate exit that the poll collected; a stage never collected does not exclude a respondent. An unanswered knowledge item within a completed questionnaire receives zero credit, but an absent or unavailable questionnaire does not receive a zero score. For the National Issues Convention, both the main analysis and the timing comparisons require observed pre-arrival and exit questionnaires. Pair-specific comparisons include attendees without known discussion groups when their attendance and questionnaire responses are documented; the common three-wave comparisons require recorded group membership. We compare pre-arrival and exit knowledge in 31 polls and all three measurements in five. Appendix C reports the available pairs and their sample sizes.

America in One Room and the climate poll interviewed separate samples who were never invited to deliberate (Fishkin et al. 2021). We compare changes in their scores with changes among participants in the main sample, using the same questions in both arms. The climate sample identifies completion of the deliberations rather than attendance at any session. These comparisons may reflect selection into attendance or completion as well as deliberation. Additional control comparisons and follow-ups appear in Appendix B.3.

5.1 Knowledge

We measure knowledge with factual questions, mostly multiple choice or true-false, along with some numerical estimates and questions about party or candidate positions (Luskin 1987; Luskin and Bullock 2005; Price 1999; Mondak 1999, 2001). Appendix D lists the items. Correct responses are scored as 1, incorrect responses and “don’t knows” as 0 (Luskin and Bullock 2011), and these scores are then averaged across items to give the proportion answered correctly.

The analysis includes 8,450 participants in 622 groups and 27 polls. The demographic model retains 8,335 participants and all 27 polls; 115 participants have missing covariates. The reading model includes 3,601 participants in 12 polls. The attitude model retains 8,310 participants in all 27 polls; 25 additional participants lack attitude measures.

5.2 Who attends?

Participation is voluntary, so participants may differ from the people invited or recruited to attend. Their initial knowledge helps characterize whom the gain estimates describe. Table 1 compares initial scores for attendees in America in One Room and completers in the climate study with the remaining recruitment or invitation sample and with people not invited to deliberate. A positive gap means that the analyzed participants started with more knowledge; it does not measure how much they learned.

Attendees in America in One Room and completers in the climate study had higher initial scores than both comparison groups. In America in One Room, the pre-arrival gaps are 8.0 and 5.3 percentage points, respectively. In the climate poll they are 10.7 and 10.1 points (Table 1). In 2019, the recruitment sample was subsequently subsampled for invitations, so we cannot identify every nonattender as someone who declined an invitation. In the climate study, baseline respondents in the treatment sample were invited to register (NORC at the University of Chicago 2019, 6; 2021, 4). These summaries use everyone with an observed baseline and the relevant attendance or completion classification, including people outside the grouped-model sample. Other climate invitees include partial attendees, people with documented zero sessions, and people whose nonattendance

is inferred from an absent post questionnaire. The completion comparison retains all other invitees. These comparisons therefore have different denominators from the paired gain and group-model samples.

The six-country antimicrobial-resistance study supplies an additional baseline comparison (Mendelson et al. 2026). Its 1,280 participants scored 4.7 percentage points above 1,139 controls before invitations. This unweighted comparison describes the survey respondents across six countries. We lack observations for the 1,847 invitees who did not attend. Discussion-group identifiers are unavailable, so this study does not enter the group models or the group-bootstrap learning comparisons.

Table 1: Pre-arrival knowledge differences: participants minus each comparison group. The climate study identifies completers; other invitees may have attended some sessions. The antimicrobial-resistance row pools six country samples without weights. N counts observed baseline scores; gaps are percentage points. Descriptive summaries have no estimated sampling intervals.

Poll	Comparator	Participant N	Comparator N	Gap (pp)
America in One Room	Uninvited controls	523	1,101	5.3
America in One Room	Recruitment nonattenders	523	2,215	8.0
America in One Room: Climate	Uninvited controls	962	834	10.1
America in One Room: Climate	Other invitees	962	7,018	10.7
Antimicrobial Resistance	Uninvited controls	1,280	1,139	4.7

The baseline gaps show that attendees began with greater knowledge; they do not by themselves establish bias in the gain estimates. Comparing changes removes fixed differences in initial knowledge. An attendee-control comparison identifies the effect of participation if, without participation, the groups would have experienced the same average change. That assumption can fail if attendance is associated with preparation or learning outside the event.

5.3 Participant and group characteristics

Education. We use the source-specific ordered categories of lower, middle, and higher education. These retain the reviewed groupings within each poll; they do not establish equivalent qualifications across countries. The lower category is the regression reference. We interact education with initial knowledge to allow their associations with gain to depend on one another.

Gender and age. Gender is coded 1 for women and 0 for men. Age enters in decades. Missing or refused responses remain missing.

Group composition and mode. Groupmates’ initial knowledge and the proportion of women exclude the participant herself. Group size counts eligible participants with both pre-arrival and exit answers, so it is a lower bound on actual attendance when members lack an interview. Mode distinguishes online from in-person deliberation.

Reading the briefing materials. Respondent-linked reading reports are available for 12 polls. They

describe reading before discussion, recalled in the post-deliberation questionnaire. We score their ordered categories from zero to one using the source-specific response scales.

Attitude extremity and disagreement. We scale baseline policy measures to 0–1 using their response endpoints. The earlier polls retain their existing policy indices; the additional polls use individual policy preferences. Extremity is the participant’s mean absolute distance from the midpoint, $E_i = J_i^{-1} \sum_j |A_{ij} - .5|$, over observed measures. Group disagreement is the average absolute difference between group members’ responses:

$$H_g = \frac{1}{J_g} \sum_{j=1}^{J_g} \frac{2}{n_{gj}(n_{gj} - 1)} \sum_{i < k} |A_{ij} - A_{kj}|.$$

Here n_{gj} counts group members with an observed response to measure j , and J_g counts measures with at least two such responses. A difference of .10 in H_g means the average pairwise gap is ten percent of the response scale larger. Reversing a scale leaves both measures unchanged. Missing responses are excluded. Policy batteries differ in content and aggregation across polls, so these measures summarize attitudes toward each poll’s own issues.

5.4 Guessing and uncertainty

A correct answer can reflect knowledge or guessing. We apply the item model of Cor and Sood (2016) separately to each poll’s paired answers. Under the assumption that knowledge is not lost between interviews, it estimates the share who learned each fact. We use this adjustment only for the pre-arrival-to-exit comparisons, not for the later follow-ups. We average across questions to obtain learning on a 0–1 scale. In each control study, we fit this model separately to attendees and controls and subtract their estimated learning. Both arms use the same items; their guessing probabilities are estimated separately. Table 3 uses observed individual gains, since the item model estimates learning rates rather than an individual latent outcome.

We estimate uncertainty using a hierarchical bootstrap. For estimates across polls, we sample polls with replacement separately within online and in-person modes, retaining their observed counts, and then sample whole discussion groups within each selected poll. For an individual poll, we hold the poll fixed and sample its groups. Independently sampled controls are single-person units, sampled separately from attendees. Each sampled participant retains all items and interviews. We refit the models and recompute denominators in each of 999 draws, using the 2.5th and 97.5th percentiles for 95% confidence intervals. Thus the same resampling scheme propagates sampling and guessing-model estimation uncertainty. Stratifying by mode preserves support for the online coefficient in every draw. Inference treats each poll’s question battery as fixed. The demographic gap comparisons use whole-poll resampling, described with Appendix Table B4.

6 How Much Do Participants Learn?

An absolute gain is the change in the percentage of items answered correctly. A relative gain divides that change by the poll’s initial mean score: an increase from 40% to 50% correct is 10 percentage points, or 25% of the initial score. We calculate relative gains from poll means rather than averaging individual ratios, which are undefined for people with zero initial scores. The main pooled estimates give each poll equal weight; the relative result is the average of the poll-specific ratios, not the ratio of two pooled means.

Across the main sample, the average absolute gain is 14.6 percentage points (95% interval 12.5 to 17.1). The average relative gain is 40.2% (95% interval 30.3 to 52.1%). Estimated learning after adjustment for guessing averages 19.4 percentage points (16.9 to 22.7). Precision weighting lowers the observed average to 12.8 points. Gains are larger in person than online, averaging 16.4 and 8.5 points, respectively (Appendix Table B2). These mode differences may also reflect differences in participants, topics, and timing; they do not identify an effect of meeting in person.

A before-and-after gain can include learning from repeated questions or events outside deliberation. Figure 1 therefore also compares attendees' gains with controls' gains. The observed differences are 13.1 percentage points in America in One Room and 9.3 percentage points in the climate poll. Their equally weighted average is 11.2 percentage points (95% interval 9.8 to 12.7), rising to 14.5 percentage points after guessing adjustment (12.9 to 16.1). Both estimates use the same attendees, controls, and questions. These comparisons do not eliminate selection into attendance.

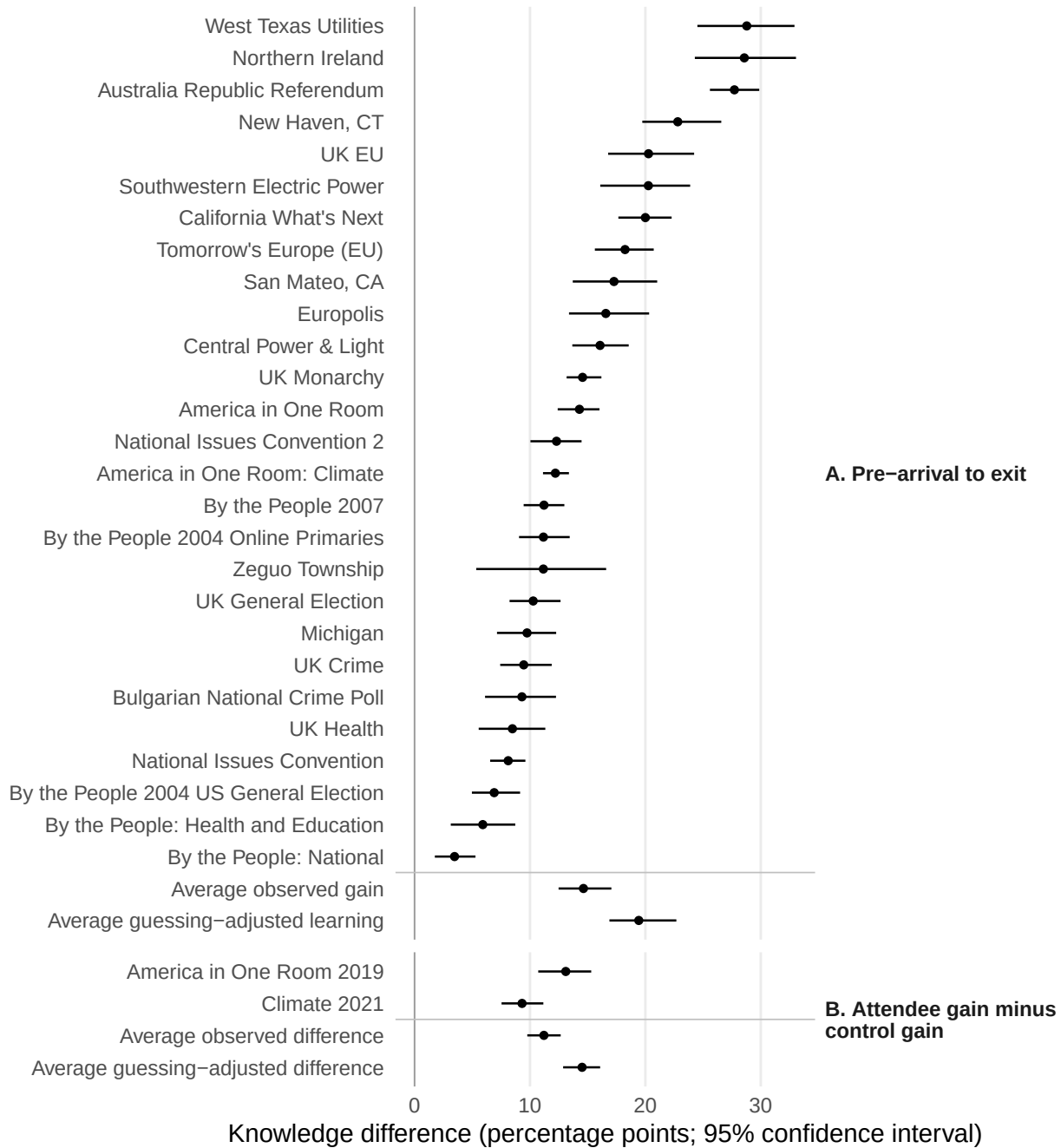


Figure 1: Knowledge-score changes in percentage points, with 95% hierarchical-bootstrap confidence intervals. A: pre-arrival to exit changes. For online polls, pre-arrival is before the discussion sessions begin. The main sample requires observed questionnaires at the stages each poll collected through immediate exit. B: observed attendee gain minus control gain in the two polls with pre-arrival and exit interviews. Each panel ends with equally weighted observed and guessing-adjusted averages. Whole discussion groups are resampled within polls; independent controls are resampled individually. Pooled intervals resample polls within mode. With one control study per mode, the pooled control interval is conditional on these two studies. Each draw refits the guessing model, treating question batteries as fixed.

6.1 Changes before arrival and during the event

Knowledge increases both before arrival and during the event in the five polls with comparable measurements at all three interviews: California, Europolis, Marousi, Michigan, and Tomorrow's Europe. Giving each poll equal weight, the mean gain is 7.6 percentage points before arrival (95% interval 5.7 to 9.5) and 6.8 points during the event (3.2 to 10.5). Figure 2 shows both components for each poll and their equally weighted means. These estimates use the same respondents at all three interviews within each poll, so the total gain equals the sum of its two components before rounding. Their common three-wave sample differs from Figure 1's pair-specific sample.

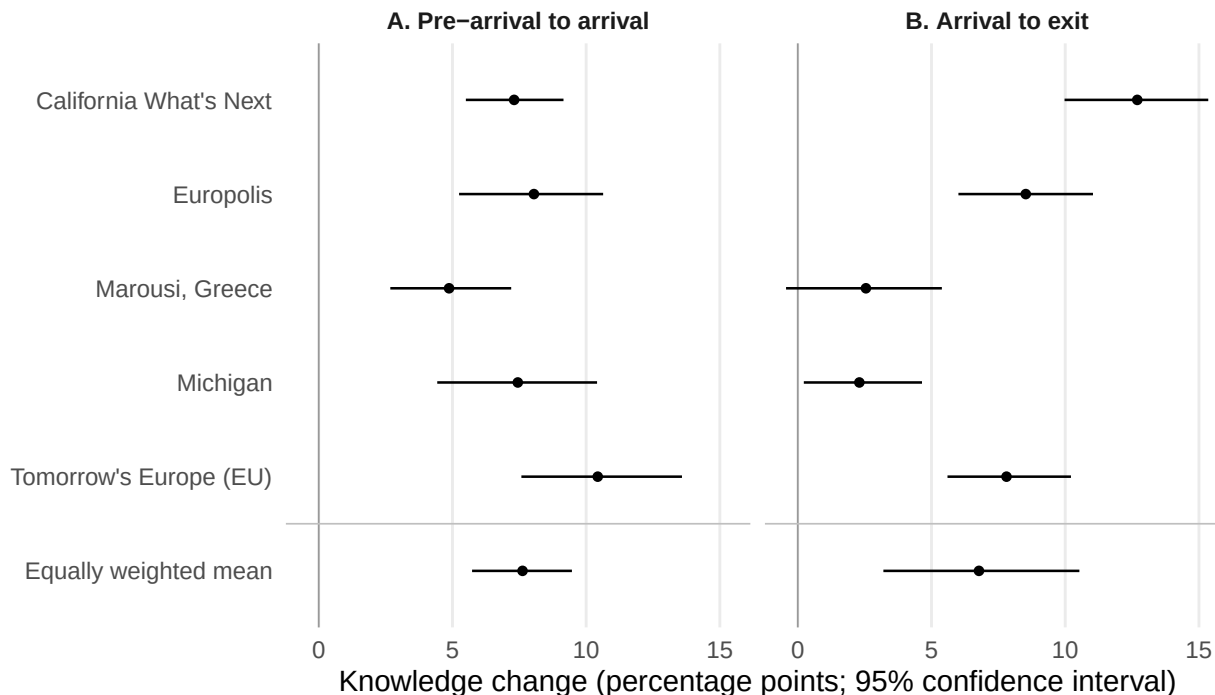


Figure 2: Knowledge changes before arrival and during the event among 1,504 attendees with all three questionnaires, the same questions and scoring denominator, and recorded discussion groups. Points are unweighted mean changes in proportion correct, expressed as percentage points; lines show 95 percent bootstrap intervals from 999 draws. Poll intervals resample discussion groups; equally weighted mean intervals resample polls and groups, retaining each person's three scores together. Both panels use the same poll order and common sample. Michigan contributes its full nine-question battery once. These samples differ from the pair-specific samples in Figure 1.

The estimates do not adjust for guessing. Michigan's main row uses its full nine-question battery; its four-question party-placement comparison remains a sensitivity in Appendix Table C1, which also reports total pre-arrival-to-exit gains and sample sizes. The Marousi comparisons exclude four otherwise eligible respondents whose discussion groups are unknown. Its original merge also contains conflicting telephone and exit identifiers; these estimates therefore depend on the original matching of respondents across interviews, which remains unresolved.

Changes before arrival could reflect briefing, preparation, repeated testing, or other events between interviews. Arrival-to-exit changes isolate the event's time window, but do not identify deliberation's causal effect without a credible comparison. Even on a common sample, these estimates

describe the attendees with observed questionnaires at all three phases; they do not describe all recruited citizens or all attendees.

6.2 How much knowledge persists?

Later interviews allow us to ask how much of the gain remains after the event. We compare pre-arrival, exit, and follow-up scores among the same attendees answering the same questions at all three interviews. Table 2 reports the National Issues Convention’s follow-up about ten months after the event and the climate poll’s follow-up about a year later (Luskin and Fishkin 2005, 18; Fishkin et al. 2025). Neither follow-up enters the main pre-arrival-to-exit estimate.

Table 2: Knowledge changes among attendees observed at all three interviews. Entries are percentage points; brackets contain 95 percent discussion-group-bootstrap intervals.

Poll and follow-up	N	Pre-arrival to exit	Exit to follow-up	Pre-arrival to follow-up
America in One Room: Climate (one year)	845	12.2 [11.0, 13.4]	-7.9 [-9.2, -6.8]	4.2 [3.1, 5.3]
National Issues Convention (ten months)	383	8.4 [6.6, 10.1]	1.4 [-0.4, 3.3]	9.8 [8.2, 11.2]

Knowledge remains above baseline at follow-up in both studies. Climate participants lose some of their initial gain after exit, whereas the National Issues Convention shows no clear decline. These comparisons describe returning attendees. Using the same people at each interview keeps changes in sample composition from being mistaken for changes in knowledge, but does not establish that those lost to follow-up had the same experience. Adjusting for observed predictors of follow-up response yields similar point estimates (Appendix C.1). Changes after exit can reflect forgetting, further learning, or both. We therefore use observed scores without the guessing model’s no-forgetting assumption. In the climate poll, the baseline-adjusted attendee-control gap also remains positive a year later (Appendix B.3).

7 Who Learns?

7.1 Who gains more?

Table 3 compares gains conditional on measured initial knowledge. The initial-knowledge coefficient in the baseline model is $-.489$ $[-.554, -.436]$. Because scores have a ceiling and contain measurement error, this should not be read as a causal effect of prior knowledge. The second column brings the education, age, and gender hypotheses into the same set of polls. The third column adds attitudes, and the fourth adds briefing reading. In both control studies, the attendee-control difference is larger among respondents who start with less knowledge. The climate poll also shows a larger difference among respondents without a bachelor’s degree, while the corresponding education estimate in the 2019 poll is imprecise (Appendix Table B5).

At the reference initial score of zero, both education categories have larger estimated gains than the lower-education group; the education-by-initial-knowledge interactions allow these gaps to vary with starting knowledge. The age estimate is $-.001$ $[-.006, .003]$ per decade. The intervals for

Table 3: Change in proportion correct (exit minus pre-arrival), conditional on initial knowledge. Coefficients with 95 percent hierarchical-bootstrap confidence intervals below.

	Baseline	Demographics	+ Attitudes	+ Reading
Intercept	.335 [.266, .403]	.269 [.195, .330]	.290 [.209, .377]	.282 [.165, .413]
Initial knowledge	-.489 [-.554, -.436]	-.504 [-.617, -.390]	-.497 [-.601, -.389]	-.499 [-.919, -.217]
Middle education		.100 [.046, .159]	.104 [.052, .160]	.133 [.008, .239]
Higher education		.197 [.143, .258]	.200 [.147, .260]	.217 [.092, .327]
Initial x middle education		-.007 [-.121, .100]	-.016 [-.123, .085]	-.101 [-.375, .330]
Initial x higher education		-.077 [-.196, .031]	-.085 [-.195, .019]	-.131 [-.429, .279]
Age (decades)		-.001 [-.006, .003]	-.001 [-.006, .003]	-.007 [-.016, .001]
Attitude extremity			.010 [-.099, .103]	
Group size	-.001 [-.003, .002]	.000 [-.002, .002]	.000 [-.002, .002]	-.001 [-.004, .003]
Groupmates' mean initial	.105 [-.016, .248]	.086 [-.021, .215]	.081 [-.020, .206]	.008 [-.183, .191]
Policy disagreement			-.079 [-.232, .075]	
Female		-.025 [-.070, .020]	-.027 [-.071, .019]	-.041 [-.114, .028]
Female groupmates (share)		-.022 [-.081, .030]	-.024 [-.082, .028]	-.052 [-.148, .026]
Female x female groupmates		.021 [-.062, .108]	.025 [-.059, .112]	.037 [-.097, .166]
Online	-.035 [-.101, .030]	-.047 [-.129, .022]	-.044 [-.125, .030]	.001 [-.066, .079]
Briefing reading				.064 [.033, .095]
Participants	8,450	8,335	8,310	3,601
Small groups	622	622	622	252
Polls	27	27	27	12
Marginal R-squared	0.274	0.325	0.326	0.315
Conditional R-squared	0.369	0.440	0.440	0.473

Notes. Groupmates' mean initial knowledge excludes the respondent herself; both that mean and group size use members with initial and later scores. All models have poll and small-group random intercepts. Education uses each source's lower, middle, and higher ordered categories; the lower category is the reference. These categories do not represent identical qualifications across polls. Marginal R^2 covers fixed effects; conditional R^2 includes random intercepts. Columns 3 and 4 separately add attitudes and reading (0–1) to column 2. Reading reports are available for fewer polls. Replacing pairwise disagreement with the mean within-group SD gives a dispersion coefficient of $-.080$ $[-.260, .105]$; the peer-knowledge coefficient is $.081$ $[-.020, .207]$.

age, gender, and the gender-by-group-composition interaction include zero. The gender coefficient compares women with men when the share of female groupmates is zero; the interaction shows how that difference changes as the share increases. The third column adds baseline attitude extremity and policy disagreement. On the same participants, adding these measures changes the peer-knowledge coefficient from .078 to .081; the intervals for both added attitude measures include zero.

Reported reading of the briefing materials is also associated with larger gains. Across 12 polls, the difference between the lowest and highest reading categories is 6.4 percentage points (95% interval 3.3 to 9.5), conditional on initial knowledge, demographics, group characteristics, and mode, with poll and small-group random intercepts. These reports were collected after deliberation, so the association need not reflect the effect of reading alone.

7.2 Do knowledge gaps narrow?

The gender gap narrows, but the educational gap does not show comparable convergence. On the same respondents before and after, the average difference between men and women declines from 7.0 to 4.6 percentage points. The reduction is 2.5 points (95% interval 1.3 to 3.8). Women gain more on average, although a gap remains at exit.

For education, relative standing within a poll is also relevant: the same qualification can occupy different positions across countries and cohorts. We therefore compare participants above their poll’s median education with those at or below it in the 20 polls with that definition. The gap increases from 9.3 to 12.2 points, widening by 2.9 points (95% interval 0.9 to 5.2). Using the source-specific ordered categories, the higher-versus-lower gap also widens. The higher-versus-middle gap changes little (Appendix Table B4). Each comparison gives eligible polls equal weight.

These are changes in observed gaps, not differences conditional on equal starting knowledge. Women begin with lower scores, so their larger average gain can coexist with the imprecise conditional gender association in Table 3. The comparisons span pre-arrival to exit and include preparation as well as the event; they do not establish that deliberation caused the gaps to narrow or widen.

7.3 Group composition and learning

Table 3 addresses the peer-knowledge hypothesis directly. A ten-percentage-point difference in groupmates’ mean initial knowledge is associated with 1.05 percentage points of additional gain in the baseline model and 0.86 after demographic adjustment. The intervals in the table quantify uncertainty about these pooled associations.

The average of the separate within-poll peer-knowledge associations is also positive but imprecise: a ten-percentage-point difference in groupmates’ initial knowledge is associated with 1.1 points of additional gain (95% interval -0.6 to 2.8). Appendix Figure B1 displays the estimates across polls. Its average gives each poll equal weight; Table 3 instead fits a common slope in a hierarchical model.

Appendix Table B6 examines whether participants with similar baseline characteristics were grouped together. For initial knowledge, the association with groupmates’ initial knowledge is .050 [-0.028, .121]. Its interval includes zero. This diagnostic concerns characteristics measured before discussion; later knowledge and gain can themselves respond to group composition. A causal interpretation of peer coefficients also requires evidence about the assignment process. Group members may finish with similar knowledge because they received the same briefing and took part

in the same discussion. Their exit scores alone cannot establish whether they learned from one another (Manski 1993). We therefore use groupmates' initial knowledge in the peer models.

8 Conclusion

Participants' factual knowledge increases between recruitment and the end of deliberation. Reporting absolute and relative gains makes the magnitude interpretable, while adjustment for guessing addresses one source of measurement error. In the two polls with common before-and-after control interviews, attendees show larger gains than controls. Interpreting the gain differences causally requires attendees and controls to have comparable changes without participation; the peer regressions are descriptive. The three-wave comparisons show that knowledge can increase before arrival as well as during the event. Generalizing beyond attendees requires assumptions about those who did not attend or complete later questionnaires. The knowledge questions concern facts selected by organizers, so these estimates describe learning about the topics the events emphasized.

Higher average knowledge does not imply that knowledge becomes more equally distributed. On average, the gender gap narrows without disappearing, while the gap between respondents classified as more and less educated within their poll widens. Deliberation and its surrounding preparation can therefore accompany both broader learning and persistent or increasing educational differences in knowledge.

The gains also need to be considered against how much room there was to learn. Giving each poll equal weight, participants answered 44.5% of questions correctly at the initial interview and 59.1% at the later interview. The gain of 14.6 percentage points closed 26.3% of the initial 55.5-point gap to a perfect score. At the later interview, 40.9% of answers were still incorrect or blank. This is a comparison of average scores, not a count of previously unknown facts learned: correct answers can reflect guessing, and net gains can conceal both learning and forgetting.

Briefing materials, moderated discussions, and opportunities to question experts give participants unusually favorable conditions for learning about the issues on which they are tested. We cannot assume that every tested fact was discussed or that every participant read the relevant material. Still, substantial gaps remain despite this intensive support, suggesting that increasing factual knowledge is difficult even when citizens have time, information, and reason to attend to it. For democratic theory, these findings support the possibility of more informed public judgment while tempering expectations of comprehensive factual knowledge. A perfect test score is not a prerequisite for sound judgment, and factual recall does not exhaust what participants may learn about arguments, tradeoffs, or other people's experiences. The normative case for deliberation should therefore consider both the knowledge it adds and how well it helps citizens make decisions with the knowledge they have.

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A The Deliberative Polls

These 27 polls supply the main learning and discussion-group sample. The comparisons by interview timing use separate samples, described in Appendix C. Three also provide control comparisons, including Northern Ireland’s T3-only contrast. Tables A1 and B1 list polls in the same chronological order.

Table A1: Polls and interview pairs used in the main analysis. t0 is pre-arrival and t2 exit.

Poll	Year	Mode	Control	Interviews
UK Crime	1994	In person	No	t0 to t2
UK EU	1995	In person	No	t0 to t2
Central Power & Light	1996	In person	No	t0 to t2
National Issues Convention	1996	In person	No	t0 to t2
Southwestern Electric Power	1996	In person	No	t0 to t2
UK Monarchy	1996	In person	No	t0 to t2
West Texas Utilities	1996	In person	No	t0 to t2
UK General Election	1997	In person	No	t0 to t2
UK Health	1998	In person	No	t0 to t2
Australia Republic Referendum	1999	In person	No	t0 to t2
Bulgarian National Crime Poll	2002	In person	No	t0 to t2
New Haven, CT	2002	In person	No	t0 to t2
By the People: National	2003	Online	No	t0 to t2
National Issues Convention 2	2003	In person	No	t0 to t2
By the People 2004 Online Primaries	2004	Online	No	t0 to t2
By the People 2004 US General Election	2004	Online	No	t0 to t2
By the People: Health and Education	2005	Online	No	t0 to t2
Zeguo Township	2005	In person	No	t0 to t2
By the People 2007	2007	Online	No	t0 to t2
Northern Ireland	2007	In person	Yes	t0 to t2
Tomorrow’s Europe (EU)	2007	In person	No	t0 to t2
San Mateo, CA	2008	In person	No	t0 to t2
Europolis	2009	In person	No	t0 to t2
Michigan	2009	In person	No	t0 to t2
California What’s Next	2011	In person	No	t0 to t2
America in One Room	2019	In person	Yes	t0 to t2
America in One Room: Climate	2021	Online	Yes	t0 to t2

B Further results

Table B1: Initial knowledge (percent correct), observed gains, and guessing-adjusted learning (percentage points) in the main sample. Interview timing and rules for including respondents with uncertain questionnaire completion vary by poll.

Poll	Initial (%)	Gain (pp)	Adjusted (pp)
UK Crime	52.2	9.5	15.0
UK EU	51.7	20.3	26.7
Central Power & Light	29.4	16.1	20.3
National Issues Convention	47.1	8.1	12.3
Southwestern Electric Power	43.5	20.3	30.1
UK Monarchy	64.8	14.6	21.7
West Texas Utilities	39.9	28.8	35.8
UK General Election	54.5	10.3	18.2
UK Health	65.9	8.5	11.7
Australia Republic Referendum	48.2	27.7	38.8
Bulgarian National Crime Poll	43.2	9.3	12.9
New Haven, CT	36.7	22.8	29.4
By the People: National	39.5	3.5	4.8
National Issues Convention 2	37.3	12.3	15.0
By the People 2004 Online Primaries	50.8	11.2	13.9
By the People 2004 US General Election	75.2	6.9	9.4
By the People: Health and Education	35.2	5.9	7.2
Zeguo Township	31.3	11.2	14.3
By the People 2007	50.7	11.2	13.2
Northern Ireland	21.8	28.6	32.8
Tomorrow's Europe (EU)	36.5	18.2	24.1
San Mateo, CA	14.5	17.3	19.8
Europolis	19.7	16.6	19.4
Michigan	34.1	9.7	12.3
California What's Next	62.0	20.0	30.5
America in One Room	45.8	14.3	18.4
America in One Room: Climate	68.6	12.2	16.7

Table B2: Average learning across polls, with hierarchical-bootstrap confidence intervals.

Quantity	Weighting	Polls	Estimate	95% interval
Observed gain (pp)	Equal	27	14.6	12.5 to 17.1
Guessing-adjusted learning (pp)	Equal	27	19.4	16.9 to 22.7
Observed gain (pp)	Precision	27	12.8	11.0 to 14.9
Guessing-adjusted learning (pp)	Precision	27	16.6	14.6 to 19.1
Observed gain: in person (pp)	Equal	21	16.4	13.6 to 19.5
Observed gain: online (pp)	Equal	6	8.5	5.8 to 10.9

Observed gain is the exit score minus the pre-arrival score; guessing-adjusted learning estimates the percentage who learned an item, averaged across questions. Equal weighting gives each poll the same weight. Precision weighting uses $w_d = 1/\widehat{\text{Var}}(\hat{\theta}_d)$, with sampling variances estimated by resampling discussion groups within each poll. The weighted average is $\sum_d w_d \hat{\theta}_d / \sum_d w_d$. These weights are held fixed when resampling polls and groups for its interval. The mode rows split the equal-poll average into in-person and online polls.

B.1 Sensitivity to zero knowledge scores

A zero score can reflect incorrect answers, explicit “don’t know” responses, or an entirely blank knowledge section on an otherwise completed questionnaire. We retain all of these cases in the main analysis. Table B3 separately removes people with a zero score at either the pre-arrival or exit interview, and people whose entire knowledge section is blank at either interview. Explicit “don’t know” responses do not count as blanks; uncertain blankness remains in the sample. These exclusions change the population being described; they are sensitivity checks, not alternative diagnoses of nonparticipation.

Table B3: Baseline model under alternative treatments of zero scores. Entries show coefficients and 95 percent intervals.

Sample	N	Initial knowledge	Groupmates’ mean initial	Online
Main	8,450	-.489 [-.554, -.436]	.105 [-.016, .248]	-.035 [-.101, .030]
Exclude zero scores	7,626	-.505 [-.569, -.458]	.062 [-.063, .202]	-.037 [-.110, .033]
Exclude blank batteries	8,344	-.487 [-.551, -.436]	.099 [-.023, .244]	-.036 [-.104, .028]

Excluding zero scores lowers the estimated peer-knowledge association; excluding wholly blank batteries changes it little. All three intervals include zero.

All three specifications use the baseline model in Table 3. Group size and groupmates’ mean knowledge are recalculated for each sample. N counts respondents in the grouped model. Intervals use 999 hierarchical bootstrap draws with the same procedure as the main analysis.

Table B4: Knowledge gaps before and after, in percentage points. Negative changes indicate narrowing. Intervals resample whole polls within mode, preserving paired interviews.

Comparison	Polls	Pre-arrival	Exit	Change [95% interval]
Men minus women	27	7.0	4.6	-2.5 [-3.8, -1.3]
Poll-specific higher minus lower education	20	9.3	12.2	2.9 [0.9, 5.2]
Higher minus lower education	26	20.6	25.7	5.1 [0.6, 9.6]
Higher minus middle education	27	11.3	11.8	0.6 [-1.1, 2.3]

These comparisons use the demographic-model sample and calculate each gap within a poll before averaging equally across polls. A poll enters a comparison only if both categories are observed. The relative-education comparison uses each poll’s reviewed ordered education measure. The median is calculated among unique historical participants in each of the 20 polls with this measure; we compare participants above the median with those at or below it. Tied categories remain together, so the two groups need not be equally large. Table 3 uses three education categories. Europolis has no respondents in the lower category in this sample, so the higher-versus-lower comparison covers 26 polls; the gender and higher-versus-middle comparisons cover 27. Some education categories contain few respondents within individual polls. For these gap comparisons, we resample whole polls within mode in 999 draws, retaining all respondents and both interviews together. The interval for the change is calculated from paired changes in each draw, not by subtracting the endpoints of separate intervals. These comparisons use the same scoring and questionnaire-completion rules as the main analysis.

Table B5: How the baseline-adjusted attendee-control difference varies with initial knowledge and education.

Study	Comparison	Difference (pp)	95% interval	N
America in One Room 2019	Initial score +10 pp	-1.5	-2.5 to -0.8	1,367
America in One Room 2019	Bachelor’s degree	2.7	-1.6 to 7.2	1,367
America in One Room: Climate 2021	Initial score +10 pp	-2.5	-3.1 to -1.9	1,633
America in One Room: Climate 2021	Bachelor’s degree	-4.7	-7.6 to -1.8	1,633

Table B5 reports changes in the attendee-control difference, not changes in everyone’s knowledge. N is the regression sample size. Degree compares a bachelor’s degree with no degree. A negative entry means the attendee advantage is smaller for people with higher initial knowledge or a bachelor’s degree. For example, the America in One Room initial-knowledge estimate of -1.5 percentage points compares two people whose starting scores differ by ten percentage points. Each regression includes attendance, initial score, the listed characteristic, and its interaction with attendance. Centering initial scores at their sample mean does not change the reported interaction. All intervals use the hierarchical bootstrap. Selection into attendance and score ceilings limit causal interpretation.

B.2 Groupmates’ knowledge of missed questions

The demographic specification can also include groupmates’ initial correctness on the particular questions the participant missed at the initial interview. For each missed question, we calculate the proportion of other group members who answered correctly, then average across those questions. This targeted measure enters alongside groupmates’ overall initial knowledge. Respondents who answered every initial question correctly receive zero on the targeted measure and a separate indicator for having no missed questions.

The targeted peer coefficient is -.093 [-.182, -.015] among 8,335 participants in 27 polls. The coefficient on overall peer knowledge is .167 [.039, .316], and the no-missed-questions indicator is -.065 [-.118, -.010]. All other regressors, random intercepts, and the bootstrap follow Table 3.

B.3 Additional control comparisons

Regressing later scores on attendance and initial scores answers a different question from comparing gains. The baseline-adjusted attendee-control differences are 14.2 percentage points in America in One Room (12.3 to 16.2), and 12.1 in the climate poll (10.5 to 13.7). These intervals use the same bootstrap as the main comparisons.

A year after the climate poll, the baseline-adjusted gap is 5.5 percentage points (3.5 to 7.2). Northern Ireland has only a follow-up control interview: 93 returning attendees scored 17.7 percentage points above 150 controls (12.3 to 23.5). This is a cross-sectional comparison without a control baseline. We do not apply the main guessing model’s no-forgetting assumption to these later outcomes.

B.4 Poll-specific peer associations

The following estimates fit a separate slope in each poll, conditional on own initial knowledge. Their equal-poll average is .111 [-.064, .282].

B.5 Baseline composition of discussion groups

We regress each participant’s baseline characteristic on its mean among the other group members, controlling for the mean among other members of the poll and poll fixed effects (Guryan et al. 2009). This is a descriptive sorting diagnostic; it does not reconstruct the original assignment mechanism. A randomization test would require the original assignment pools and restrictions, including any scheduling or language constraints. Balance on observed covariates alone does not establish random assignment.

Table B6: Baseline composition of discussion groups. Associations with 95 percent hierarchical-bootstrap intervals.

Characteristic	Association	Participants
Initial knowledge	.050 [-.028, .121]	8,450
Female	-.001 [-.059, .053]	8,439
Age	-.006 [-.061, .045]	8,403
Higher education category	.046 [-.018, .103]	8,376

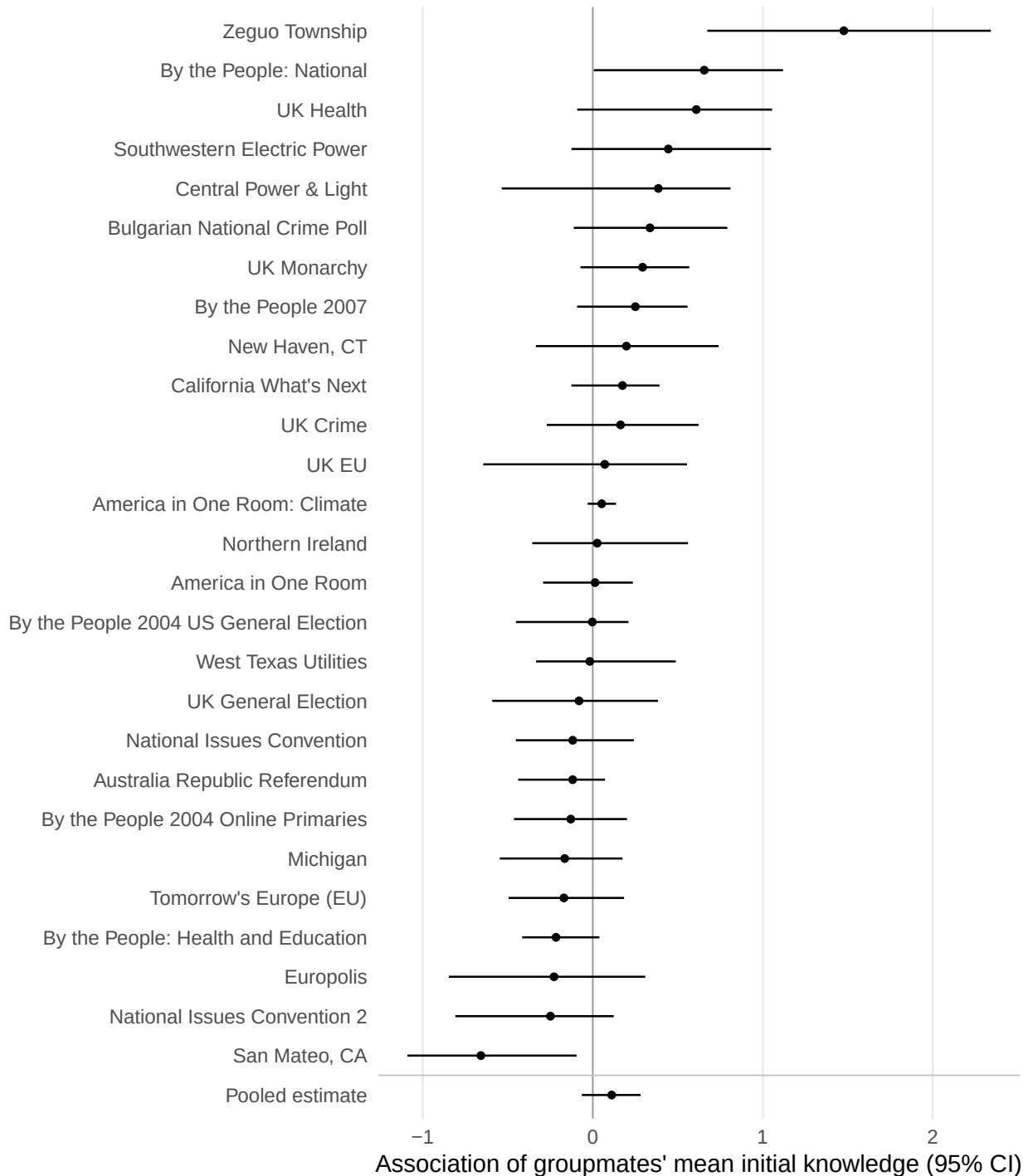


Figure B1: Association between the other group members' mean initial knowledge and own knowledge gain, by poll, with 95% intervals. Each estimate controls for own initial knowledge. Intervals use the hierarchical bootstrap. The pooled estimate weights polls equally and resamples polls within mode and then their groups. These are descriptive associations.

C Interview timing and analysis samples

Table C1 reports all three changes on the same three-wave sample within each poll, including the four-question Michigan sensitivity. Figure 2 uses one primary battery per poll and reports the equally weighted means of its two components. Michigan’s nine-question primary battery contains five factual questions and four party-placement questions at every interview; the four-question sensitivity never supplies a second Michigan observation to those means. The total pre-arrival-to-exit change is the sum of the two component changes before rounding.

Table C1: Knowledge changes among attendees with all three questionnaires and recorded discussion groups. Entries are percentage points; brackets contain 95 percent group-bootstrap intervals. Items reports the battery size where documented; a dash marks an unavailable count.

Poll	Items	N	Pre-arrival to arrival	Arrival to exit	Pre-arrival to exit
California What’s Next	5	386	7.3 [5.5, 9.2]	12.7 [10.0, 15.3]	20.0 [17.7, 22.5]
Europolis	6	348	8.0 [5.2, 10.6]	8.5 [6.0, 11.0]	16.6 [13.3, 20.1]
Marousi, Greece	7	129	4.9 [2.7, 7.2]	2.5 [-0.4, 5.4]	7.4 [5.2, 9.8]
Michigan	9	309	7.4 [4.4, 10.4]	2.3 [0.2, 4.6]	9.7 [7.3, 12.5]
Michigan	4	309	10.8 [7.4, 14.4]	1.5 [-2.4, 5.0]	12.4 [8.3, 16.5]
Tomorrow’s Europe (EU)	11	332	10.4 [7.6, 13.6]	7.8 [5.6, 10.2]	18.2 [15.6, 20.7]

Questionnaire completion creates an additional selection stage. Appendix Table C3 compares baseline knowledge among known attendees with observed, absent, or uncertain exit questionnaires. A blank quiz within an observed questionnaire scores zero; a wholly absent questionnaire remains unmeasured. Respondents whose questionnaire completion is uncertain are excluded from the paired comparisons. Those whose attendance is unknown remain in the baseline summaries and sample counts, with attendance recorded as unknown. Attendance classifications use source indicators, session records, onsite questionnaires, and verified historical identity links. For the harmonized participation measure, a wholly unanswered immediate post questionnaire implies nonattendance even when a roster or session record indicates attendance; the original indicator remains available. Main analyses also require observed questionnaires at earlier stages that the poll collected. A stage that was never collected does not exclude anyone. Cases without sufficient source evidence remain unknown. Uncertain attendance does not establish that a respondent was absent, and missing scores do not establish that a poll lacked an arrival survey.

Table C2 compares knowledge among attendees who answered both questionnaires in each pair of interviews. These estimates are unweighted changes in proportion correct without guessing adjustment. The overlapping historical Primaries subset is omitted in favor of the attendee sample used in the main analysis. We use one source copy where original respondent identities, answers, scores, attendance, and discussion groups agree. We retain comparisons with different question batteries or respondent samples. For Tomorrow’s Europe, Archive and Cor–Sood replication iden-

tify overlapping source versions whose discussion-group assignments differ; their estimates are not independent polls. Each row requires observed questionnaires and the same battery and denominator at both phases; it does not require the third phase or discussion-group membership. Thus the three rows for a poll need not use the same people and need not add up. Table C1 instead restricts all three changes to the common three-wave sample with recorded discussion groups. The Items column distinguishes separate batteries: California has five common questions and an expanded eight-question arrival/exit battery; Europolis has six and nine, respectively. Michigan has nine questions across all three interviews, including four party-placement questions; a separate battery has six party placements at arrival and exit. We report descriptive estimates without intervals when discussion-group membership is unknown or too few groups are available to estimate uncertainty.

Table C2: Knowledge changes among attendees who answered both questionnaires. Changes and 95 percent group-bootstrap intervals are percentage points. Items reports battery size; a dash means an unavailable count. A missing interval means groups are unknown or too few.

Poll	Comparison	Items	N	Change (pp)	95% interval
America in One Room	Pre-arrival to exit	7	523	14.3	12.5 to 16.2
America in One Room: Climate	Pre-arrival to exit	8	962	12.2	11.1 to 13.4
Antimicrobial Resistance	Pre-arrival to exit	6	1,280	12.5	–
Australia Republic Referendum	Pre-arrival to exit	10	347	31.6	29.0 to 34.2
Australia Republic Referendum	Pre-arrival to exit	12	347	27.7	25.4 to 29.9
Bulgarian National Crime Poll	Pre-arrival to exit	7	278	9.3	6.4 to 12.4
By the People 2004 Online Primaries	Pre-arrival to exit	7	239	11.1	–
By the People 2004 US General Election	Pre-arrival to exit	9	246	7.0	4.9 to 9.4
By the People 2004 US General Election	Pre-arrival to exit	9	248	6.9	4.8 to 9.3
By the People 2007	Pre-arrival to exit	8	301	11.2	9.5 to 13.1
By the People: Health and Education	Pre-arrival to exit	6	454	5.9	3.1 to 8.6
By the People: National	Pre-arrival to exit	11	244	3.5	1.8 to 5.3
California What’s Next	Pre-arrival to arrival	5	386	7.3	5.5 to 9.2
California What’s Next	Arrival to exit	5	396	12.7	10.1 to 15.3
California What’s Next	Arrival to exit	8	396	16.7	14.7 to 18.8
California What’s Next	Pre-arrival to exit	5	386	20.0	17.7 to 22.5
Central Power & Light	Pre-arrival to exit	7	216	16.1	13.7 to 18.7
Denmark: Euro	Pre-arrival to arrival	9	358	12.6	–
Denmark: Euro	Arrival to exit	9	358	9.5	–
Denmark: Euro	Pre-arrival to exit	9	359	22.0	–
Europolis	Pre-arrival to arrival	6	348	8.0	5.2 to 10.6
Europolis	Arrival to exit	6	348	8.5	6.0 to 11.0
Europolis	Arrival to exit	9	348	8.2	6.8 to 9.6
Europolis	Pre-arrival to exit	6	348	16.6	13.3 to 20.1

Marousi, Greece	Pre-arrival to arrival	7	133	5.0	—
Marousi, Greece	Arrival to exit	7	133	2.3	—
Marousi, Greece	Pre-arrival to exit	7	138	6.7	—
Michigan	Pre-arrival to arrival	9	309	7.4	4.4 to 10.4
Michigan	Pre-arrival to arrival	4	309	10.8	7.4 to 14.4
Michigan	Arrival to exit	9	309	2.3	0.2 to 4.6
Michigan	Arrival to exit	4	309	1.5	-2.4 to 5.0
Michigan	Arrival to exit	6	309	0.4	-3.1 to 3.7
Michigan	Pre-arrival to exit	9	310	9.8	7.4 to 12.5
Michigan	Pre-arrival to exit	4	310	12.4	8.2 to 16.6
National Issues Convention	Pre-arrival to exit	8	456	9.1	7.4 to 10.8
National Issues Convention	Pre-arrival to exit	11	456	8.1	6.5 to 9.8
National Issues Convention 2	Pre-arrival to exit	11	340	12.3	10.2 to 14.3
New Haven, CT	Pre-arrival to exit	8	131	22.8	19.4 to 26.2
Northern Ireland	Pre-arrival to exit	7	124	28.6	24.2 to 33.2
San Mateo, CA	Pre-arrival to exit	8	238	17.3	13.4 to 20.9
Southwestern Electric Power	Pre-arrival to exit	5	232	20.3	16.6 to 24.1
Tomorrow’s Europe (EU): Archive	Pre-arrival to arrival	11	332	10.4	7.6 to 13.6
Tomorrow’s Europe (EU): Archive	Arrival to exit	11	332	7.8	5.6 to 10.2
Tomorrow’s Europe (EU): Cor–Sood replication	Pre-arrival to exit	11	335	18.2	15.9 to 20.7
Tomorrow’s Europe (EU): Archive	Pre-arrival to exit	11	335	18.2	15.6 to 20.7
UK Crime	Pre-arrival to exit	7	299	9.5	7.4 to 11.8
UK EU	Pre-arrival to exit	5	224	20.3	—
UK General Election	Pre-arrival to exit	15	275	10.3	8.2 to 12.3
UK Health	Pre-arrival to exit	6	228	8.5	5.5 to 11.5
UK Monarchy	Pre-arrival to exit	8	258	15.0	13.5 to 16.7
UK Monarchy	Pre-arrival to exit	9	258	14.6	13.2 to 16.3
Vermont Energy	Pre-arrival to arrival	9	146	24.4	—
Vermont Energy	Arrival to exit	9	146	11.8	—
Vermont Energy	Pre-arrival to exit	9	146	36.2	—
West Texas Utilities	Pre-arrival to exit	5	230	28.8	24.6 to 32.9
Zeguo Township	Pre-arrival to exit	4	235	11.3	—

Table C3 reports exit-questionnaire availability and baseline knowledge for Marousi, Tomorrow’s Europe, and the two control studies. An absent exit requires evidence of a wholly absent questionnaire; an uncertain exit is not recoded as absent. Baseline means use the scored records counted in the last-but-one column. The groups shown exhaust known attendees in these source frames, including people excluded from paired-change estimates.

Table C3: Exit availability among known attendees: sample counts and pre-arrival knowledge (percent correct).

Poll	Exit	Attendee N	Baseline N	Baseline (%)
America in One Room	Observed	523	523	45.8
America in One Room: Climate	Observed	962	962	68.6
Marousi, Greece	Observed	138	138	36.0
Tomorrow’s Europe (EU): Cor–Sood replication	Observed	335	335	36.5
Tomorrow’s Europe (EU): Archive	Observed	335	335	36.5

Marousi’s full telephone frame contains 1,275 records. The source attendance indicators and questionnaire records are retained to study selection and attrition; the harmonized attendance measure follows the completion rule above. An absent arrival or exit form does not establish whether someone was invited or declined an invitation. Some Cor–Sood cohorts retain unresolved attendance, recorded separately from questionnaire presence. Matching respondent numbers in separate datasets are not sufficient to establish that records describe the same person.

C.1 Adjusting for missing interviews

The comparisons above describe people who answered the relevant questionnaires. If returning respondents learned more or less than other members of the original source cohort, their gains need not describe that larger cohort. We examine this possibility in Marousi, the National Issues Convention, and the climate poll. In Marousi, the target is the original source cohort with positive attendance evidence and a pre-arrival score. This includes 21 people without an exit questionnaire who remain nonparticipants under the main analysis’s completion rule. Defining this attrition target before requiring an exit questionnaire avoids excluding the missing outcomes we seek to assess. For the two follow-up studies, the target is attendees with both pre-arrival and exit scores, including those missing at follow-up. These latter comparisons address retention among attendees whose initial gain is observed; they do not adjust for missing baseline or exit interviews.

Table C4: Observed knowledge scores within each target cohort. A dash denotes a phase outside that comparison.

Poll	Target N	Pre-arrival	Arrival	Exit	Follow-up
America in One Room: Climate	962	962	–	962	845
Marousi, Greece	159	159	154	138	–
National Issues Convention	456	456	–	456	383

Missing interviews are treated separately at each stage. A missed arrival interview does not imply that the exit interview is missing. People with unknown attendance are not classified as nonattendees, and an absent questionnaire is not assigned a knowledge score of zero.

Within each poll, we estimate the probability of observing a score using logistic regression and its conditional mean using fractional logistic regression. We combine the two predictions in an augmented inverse-probability-weighted estimate (Seaman and Vansteelandt 2018). Let R_{it} indicate an observed score at stage t , $\hat{p}_t(X_i)$ its estimated response probability, and $\hat{m}_t(X_i)$ its predicted value. The estimated mean for the target cohort is

$$\hat{\mu}_t = \frac{1}{N} \sum_{i=1}^N \left[\hat{m}_t(X_i) + \frac{R_{it}}{\hat{p}_t(X_i)} \{K_{it} - \hat{m}_t(X_i)\} \right],$$

where the residual term is zero when the score is unobserved. We calculate changes by subtracting stage means for the same target cohort, so the component changes sum to the total. Table C5 compares these estimates with complete-case changes and normalized inverse-probability-weighted changes.

The Marousi models use pre-arrival knowledge. The National Issues Convention models use pre-arrival and exit knowledge. The climate models also include age, a bachelor’s-degree indicator, and gender. We use demographics only where they are recorded for the entire target cohort, avoiding further exclusions or demographic imputation. All models are fitted separately by poll, with the same predictors in the response and outcome models.

Table C5: Knowledge changes before and after adjustment for interview nonresponse, in percentage points. Entries are point estimates; this comparison does not estimate sampling intervals.

Poll	Comparison	Complete cases	Weighted	Augmented
America in One Room: Climate	Pre-arrival to exit	12.2	12.2	12.2
America in One Room: Climate	Exit to follow-up	-7.9	-8.0	-8.0
America in One Room: Climate	Pre-arrival to follow-up	4.2	4.2	4.2
Marousi, Greece	Pre-arrival to arrival	5.0	4.5	4.5
Marousi, Greece	Arrival to exit	2.3	2.3	2.3
Marousi, Greece	Pre-arrival to exit	7.3	6.8	6.8
National Issues Convention	Pre-arrival to exit	8.4	8.1	8.1
National Issues Convention	Exit to follow-up	1.4	1.8	1.8
National Issues Convention	Pre-arrival to follow-up	9.8	10.0	9.9

The Marousi complete-case column includes 133 attendees, including the four with unknown groups excluded from Table C1. The adjusted estimates target all 159 baseline-observed members of the original source cohort. In the follow-up studies, the complete-case samples match Table 2, while adjustment targets the larger cohorts in Table C4.

The fitted response probabilities are no lower than 71.9%, and no respondent receives an inverse-probability weight above 1.39. We do not trim weights. Adjustment changes these point estimates

by at most 0.6 percentage points. This similarity concerns selection on the measured predictors. Identification still requires that, conditional on those predictors, respondents and nonrespondents have the same expected knowledge and a positive chance of response. Under that assumption, the augmented estimator is consistent if either the response model or the outcome model is correctly specified. It does not remove selection on unmeasured knowledge, establish a causal effect of attendance, or estimate what nonattenders would have learned.

D Knowledge items

The catalog lists 222 questions for the same 27 polls used in the main learning and group-model analysis (Sood 2026). Wording is paraphrased where only source labels survive; capitalization and unambiguous abbreviations are normalized for readability. Original labels and scoring rules are preserved in the cited dataset. Choices are transcribed where recoverable; truncated or unavailable options are identified. Notes explain open-response coding.

D.1 UK Crime (1994)

- **Can British courts sentence a murderer to death?** (true/false). Choices: True | False. Scored correct: False [0].
- **Does Britain have the largest prison population in Western Europe?** (true/false). Choices: True | False. Scored correct: True [1].
- **Does Britain have more prisoners serving life sentences than the rest of the European Community combined?** (true/false). Choices: True | False. Scored correct: True [1].
- **Can a defendant be tried by jury in a local magistrates court?** (true/false). Choices: True | False. Scored correct: False [0].
- **Is the number of members of Parliament about 100?** (true/false). Choices: True | False. Scored correct: False [0].
- **Is the longest permitted interval between general elections four years?** (true/false). Choices: True | False. Scored correct: False [0].
- **Is the British electoral system based on proportional representation?** (true/false). Choices: True | False. Scored correct: False [0].

D.2 UK EU (1995)

- **Has the European Union recently expanded to 15 members?** (true/false). Choices: True | False. Scored correct: True [1].
- **Is Switzerland about to join the European Union?** (true/false). Choices: True | False. Scored correct: False [2].
- **Are British income tax rates decided in Brussels?** (true/false). Choices: True | False. Scored correct: False [2].
- **Are European Parliament elections held every five years?** (true/false). Choices: True | False. Scored correct: True [1].
- **Are the Liberal Democrats the least favorable to the European Union of the three main British parties?** (true/false). Choices: True | False. Scored correct: False [2].

D.3 Central Power & Light (1996)

- **Which energy source produces the greatest amount of CPL electricity?** (multiple choice). Choices: 1: Coal | 2: Wind | 3: Natural Gas | 4: Fuel Oil | 5: Nuclear | 6: Solar | 99: dk. Scored correct: Natural Gas [3].

- **Which customer group uses the most CPL electricity?** (multiple choice). Choices: 1: Residential | 2: Commerical | 3: Industrial | 99: dk. Scored correct: Industrial [3].
- **Which CPL customer group pays the highest electric rates?** (multiple choice). Choices: 1: Residential | 2: Commerical | 3: Industrial | 99: dk. Scored correct: Commerical [2].
- **Which electricity source produces the most air emissions?** (multiple choice). Choices: 1: Nuclear | 2: Coal | 3: Natural Gas | 99: dk. Scored correct: Coal [2].
- **Which agency sets electric rates in Texas?** (open coded). Scored correct: Puc [1]. Open answer coded into source categories; code 1 (Public Utility Commission) scored correct. Other codes scored incorrect.
- **How much profit do Texas electric utilities make on fuel?** (multiple choice). Choices: 0: 0% | 1: 10% | 2: 20% | 3: 30% | 99: dk. Scored correct: 0% [0].
- **How much of the electric bill pays for fuel?** (multiple choice). Choices: 1: less than 20% | 2: 21-30% | 3: 31-40% | 4: 41-50% | 5: More Than 50% | 99: dk. Scored correct: 21-30% [2].

D.4 National Issues Convention (1996)

- **About what percentage of American children are born outside marriage?** (open numeric). Scored correct: 25-40 inclusive [25-40 inclusive]. Numeric response scored correct when within the inclusive interval shown in the answer key; missing and out-of-range responses scored incorrect.
- **About what percentage of children live in single-parent households?** (open numeric). Scored correct: 1-10 inclusive [1-10 inclusive]. Numeric response scored correct when within the inclusive interval shown in the answer key; missing and out-of-range responses scored incorrect.
- **What was the unemployment rate over the previous ten years?** (open numeric). Scored correct: 5-10 inclusive [5-10 inclusive]. Numeric response scored correct when within the inclusive interval shown in the answer key; missing and out-of-range responses scored incorrect.
- **Which listed purpose does the United States spend the most on?** (multiple choice). Choices: 1: The space program | 2: Foreign aid | 3: National defense | 4: Agriculture subsidies | 8: Do not know. Scored correct: National defense [3].
- **Which listed country does the United States trade with most?** (multiple choice). Choices: 1: Canada | 2: Japan | 3: Germany | 4: Mexico | 8: Do not know. Scored correct: Canada [1].
- **Had the United States sent ground troops to Iraq in the previous ten years?** (yes/no). Choices: Yes | No. Scored correct: Yes [1].
- **Had the United States sent ground troops to Somalia in the previous ten years?** (yes/no). Choices: Yes | No. Scored correct: Yes [1].
- **Had the United States sent ground troops to Rwanda in the previous ten years?** (yes/no). Choices: Yes | No. Scored correct: No [0].

- **Had the United States sent ground troops to Bosnia in the previous ten years?** (yes/no). Choices: Yes | No. Scored correct: Yes [1].
- **Where were Republicans on the liberal to conservative scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Slightly conservative | Conservative | Extremely conservative [5|6|7].
- **Where were Democrats on the liberal to conservative scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Extremely liberal | Liberal | Slightly liberal [1|2|3].

D.5 Southwestern Electric Power (1996)

- **Which energy source produces the greatest amount of SWEPCO electricity?** (multiple choice). Choices: 1: Coal | 2: Wind | 3: Natural Gas | 4: Fuel Oil | 5: Nuclear | 6: Solar | 99: dk. Scored correct: Coal [1].
- **Which customer group uses the most SWEPCO electricity?** (multiple choice). Choices: 1: Residential | 2: Commerical | 3: Industrial | 99: dk. Scored correct: Industrial [3].
- **Which SWEPCO customer group pays the highest electric rates?** (multiple choice). Choices: 1: Residential | 2: Commerical | 3: Industrial | 99: dk. Scored correct: Residential [1].
- **Which electricity source produces the most air emissions?** (multiple choice). Choices: 1: Nuclear | 2: Coal | 3: Natural Gas | 99: dk. Scored correct: Coal [2].
- **Which agency sets electric rates in Texas?** (open coded). Scored correct: Puc [1]. Open answer coded into source categories; code 1 (Public Utility Commission) scored correct. Other codes scored incorrect.

D.6 UK Monarchy (1996)

- **Is Princess Anne next in line to the throne?** (true/false). Choices: True | False. Scored correct: 2 false [2].
- **Is the prime minister, rather than the Queen, Britain's head of state?** (true/false). Choices: True | False. Scored correct: 2 false [2].
- **Is the Queen head of the Commonwealth?** (true/false). Choices: True | False. Scored correct: 1 true [1].
- **Is it the Queen's duty to decide the date of a general election?** (true/false). Choices: True | False. Scored correct: 2 false [2].
- **Does a prime minister take office only after being asked by the Queen?** (true/false). Choices: True | False. Scored correct: 1 true [1].
- **Is the Queen head of the Church of England?** (true/false). Choices: True | False. Scored correct: 1 true [1].
- **Is the Queen head of the Church of Scotland?** (true/false). Choices: True | False. Scored correct: 2 false [2].

- **Does Britain have an unwritten constitution?** (true/false). Choices: True | False. Scored correct: 1 true [1].
- **If a British monarch dies, who succeeds them?** (multiple choice). Choices: 1: 1 brother or siste | 2: 2 eldest brother | 3: 3 eldest sister | 4: 4 eldest child | 5: 5 eldest son | 6: 6 eldest daughter | 7: 7 none of these | 8: 8 can't choose | 9: 9 not answered. Scored correct: 5 eldest son [5].

D.7 West Texas Utilities (1996)

- **Which energy source produces the greatest amount of WTU electricity?** (multiple choice). Choices: 1: Coal | 2: Wind | 3: Natural Gas | 4: Fuel Oil | 5: Nuclear | 6: Solar | 99: dk. Scored correct: Natural Gas [3].
- **Which customer group uses the most WTU electricity?** (multiple choice). Choices: 1: Residential | 2: Commerical | 3: Industrial | 99: dk. Scored correct: Residential [1].
- **Which WTU customer group pays the highest electric rates?** (multiple choice). Choices: 1: Residential | 2: Commerical | 3: Industrial | 99: dk. Scored correct: Residential [1].
- **Which electricity source produces the most air emissions?** (multiple choice). Choices: 1: Nuclear | 2: Coal | 3: Natural Gas | 99: dk. Scored correct: Coal [2].
- **Which agency sets electric rates in Texas?** (open coded). Scored correct: Puc [1]. Open answer coded into source categories; code 1 (Public Utility Commission) scored correct. Other codes scored incorrect.

D.8 UK General Election (1997)

- **Had inflation been below five percent over the previous few years?** (true/false). Choices: True | False. Scored correct: True [1].
- **Did the Bank of England decide interest rates?** (true/false). Choices: True | False. Scored correct: False [2].
- **Was unemployment in the United Kingdom higher than in Germany?** (true/false). Choices: True | False. Scored correct: False [2].
- **Where was the Conservative Party on the income redistribution scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Gov Do Nothing to Make Inc. = | | [1|2|3].
- **Where was the Conservative Party on the tax and public services scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Gov Spend Less, Cut Taxes Lot | | [1|2|3].
- **Where was the Conservative Party on the minimum wage scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Yes Min. Wage | | [1|2|3].
- **Where was the Conservative Party on the European Union scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Keep Distance from E.U. | | [1|2|3].

- **Where was the Labour Party on the income redistribution scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: | | Gov Try Hard Make Inc. = [5|6|7].
- **Where was the Labour Party on the tax and public services scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: | | Gov Spend Lot, Tax lot [5|6|7].
- **Where was the Labour Party on the minimum wage scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Positions 5, 6, 7 [5|6|7].
- **Where was the Labour Party on the European Union scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: | | Unite with E.U. [5|6|7].
- **Where were the Liberal Democrats on the income redistribution scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: | | Gov Try Hard Make Inc. = [5|6|7].
- **Where were the Liberal Democrats on the tax and public services scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: | | Gov Spend Lot, Tax lot [5|6|7].
- **Where were the Liberal Democrats on the minimum wage scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Positions 5, 6, 7 [5|6|7].
- **Where were the Liberal Democrats on the European Union scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: | | Unite with E.U. [5|6|7].

D.9 UK Health (1998)

- **Was the standard National Health Service prescription charge ten pounds?** (true/false). Choices: True | False. Scored correct: false [0].
- **Was the proportion of older people in Britain increasing?** (true/false). Choices: True | False. Scored correct: true [1].
- **Had inflation-adjusted National Health Service spending doubled in the previous 20 years?** (true/false). Choices: True | False. Scored correct: true [1].
- **Did most people use private medical treatment instead of the National Health Service?** (true/false). Choices: True | False. Scored correct: false [0].
- **All British women can get free breast cancer screening on the NHS.** (true/false). Choices: True | False. Scored correct: false [0]. UKH-12: use exact universal availability wording; codebook opening caveat distinguishes literal false from anticipated true. Maintained false key is preserved; printed correctness counts conflict with the maintained SAV. No rekeying authorized.
- **Was cosmetic surgery never available through the National Health Service?** (true/false). Choices: True | False. Scored correct: false [0].

D.10 Australia Republic Referendum (1999)

- **How was the Governor-General appointed under the existing monarchy?** (multiple choice). Choices: 1: Chooses Governor-General | 2: No role | 3: Appointed on recommendation of Parliament | 4: Appointed on advice of prime minister (correct) | 96: None | 97: Don't know. Scored correct: Appointed on advice of prime minister (correct) [4].
- **Could the prime minister remove the proposed president?** (multiple choice). Choices: 1: Anytime without report to House | 2: Anytime with later House approval (correct) | 3: After fair trial in Parliament | 4: Cannot remove | 96: None | 97: Don't know. Scored correct: Anytime with later House approval (correct) [2].
- **Could the Governor-General dismiss the government?** (multiple choice). Choices: 1: Controls government | 2: Acts on Queen's instructions | 3: Performs only ceremonial duties | 4: Can decide to dismiss government (correct) | 96: None | 97: Don't know. Scored correct: Can decide to dismiss government (correct) [4].
- **Which existing office would the proposed president most resemble?** (multiple choice). Choices: 1: British prime minister | 2: Australian prime minister | 3: American president | 4: Governor-General (correct) | 96: None | 97: Don't know. Scored correct: Governor-General (correct) [4].
- **Which party was more concerned with social and welfare issues?** (multiple choice). Choices: 1: Liberal Party more | 2: Equally concerned | 3: Labor Party more (correct) | 96: None | 97: Not much impression/don't know. Scored correct: Labor Party more (correct) [3].
- **Which party was closer to business?** (multiple choice). Choices: 1: Liberal Party more (correct) | 2: Equally close | 3: Labor Party more | 96: None | 97: Not much impression/don't know. Scored correct: Liberal Party more (correct) [1].
- **What public office did Aden Ridgeway hold?** (multiple choice). Choices: 1: Leader of federal opposition | 2: Leader of Democrats | 3: Justice of the High Court | 4: Aboriginal senator in Parliament (correct) | 5: None of these | 6: Don't know | 97: Don't know. Scored correct: Aboriginal senator in Parliament (correct) [4].
- **What public office did Jennie George hold?** (multiple choice). Choices: 1: Secretary of Teachers Federation | 2: Labor member of federal Parliament | 3: Leader of workers' union | 4: President of actu (correct) | 5: None of these | 97: Don't know. Scored correct: President of actu (correct) [4].
- **Would the Australian flag change under the proposed republic?** (multiple choice). Choices: 1: No | 2: Yes. Scored correct: No [1].
- **Would the national anthem change under the proposed republic?** (multiple choice). Choices: 1: No | 2: Yes. Scored correct: No [1].
- **Would use of the word Royal in military service names change under the proposed republic?** (multiple choice). Choices: 1: No | 2: Yes. Scored correct: Yes [2].
- **Would Australian participation in the Commonwealth Games change under the proposed republic?** (multiple choice). Choices: 1: No | 2: Yes. Scored correct: No [1].

D.11 Bulgarian National Crime Poll (2002)

- **Could police detain a citizen for questioning at any hour without allowing refusal?** (true/false). Choices: True | False. Scored correct: False [2].
- **Could a court detain a suspect indefinitely while proving guilt?** (true/false). Choices: True | False. Scored correct: False [2].
- **Could prosecutors detain a suspect until a case was resolved?** (true/false). Choices: True | False. Scored correct: False [2].
- **Was the chief prosecutor accountable to parliament?** (true/false). Choices: True | False. Scored correct: False [2].
- **Were police forbidden to use violence against a detainee even if the person was guilty?** (true/false). Choices: True | False. Scored correct: True [1].
- **Could police detain a person for 72 hours?** (true/false). Choices: True | False. Scored correct: False [2].
- **Was a confession alone insufficient to establish guilt?** (true/false). Choices: True | False. Scored correct: True [1].

D.12 New Haven, CT (2002)

- **Which population estimate is closest to that of the greater New Haven region?** (multiple choice). Choices: 1: 250,000 | 2: 350,000 | 3: 550,000 | 4: 750,000. Scored correct: 550,000 [3].
- **Was New Haven regional job growth in the 1990s greater than, similar to, or less than the rest of the United States?** (multiple choice). Choices: 1: More | 2: About the same | 3: Less. Scored correct: About the same [2].
- **What is the main source of revenue for most town governments in greater New Haven?** (multiple choice). Choices: 1: Sales taxes | 2: Property taxes | 3: State subsidies | 4: Federal subsidies. Scored correct: Property taxes [2].
- **Did the population of New Haven city increase, stay the same, or decrease in the 1990s?** (multiple choice). Choices: 1: Increased | 2: Stayed the same | 3: Decreased. Scored correct: Decreased [3].
- **Does Connecticut law allow communities to share property tax revenues?** (multiple choice). Choices: 1: Yes | 2: No. Scored correct: Yes [1].
- **Do communities with the most valuable property tend to have the lowest, average, or highest property tax rates?** (multiple choice). Choices: 1: Lowest | 2: Average | 3: Highest. Scored correct: Lowest [1].
- **How does the Federal Aviation Authority classify Tweed New Haven Airport?** (multiple choice). Choices: 1: Major hub | 2: Medium hub | 3: Minor hub | 4: Non-hub. Scored correct: Non-hub [4].
- **Would maintaining Tweed New Haven Airport at its current level of service require significant investment?** (multiple choice). Choices: 1: Yes | 2: No. Scored correct: Yes [1].

D.13 By the People: National (2003)

- **Where did the Democratic Party stand on a 0–10 scale from doing nothing about global warming to acting now?** (placement scale). Choices: 0–10 scale: 0 = do nothing | 5 = midway | 10 = act now. Scored correct: Above midpoint [6|7|8|9|10].
- **Where did the Republican Party stand on a 0–10 scale from doing nothing about global warming to acting now?** (placement scale). Choices: 0–10 scale: 0 = do nothing | 5 = midway | 10 = act now. Scored correct: Below midpoint [0|1|2|3|4].
- **Did President Bush want to increase, decrease, or maintain foreign aid?** (multiple choice). Choices: 1: increase | 2: decrease | 3: keep it the same | 4: haven't heard anything about this. Scored correct: increase [1].
- **Did President Bush support recent international agreements to control greenhouse gases?** (multiple choice). Choices: 1: supports | 2: opposes | 3: haven't heard anything about this. Scored correct: opposes [2].
- **How much of each \$100 in the federal budget went to military spending?** (multiple choice). Choices: 1: 1 dollar, | 2: 5 dollars, | 3: 10 dollars, | 4: 20 dollars, | 5: 30 dollars or more, or | 6: don't you have much idea about that?. Scored correct: 20 dollars, [4].
- **How much of each \$100 in the federal budget went to foreign aid?** (multiple choice). Choices: 1: 1 dollar, | 2: 5 dollars, | 3: 10 dollars, | 4: 20 dollars, | 5: 30 dollars or more, or | 6: don't you have much idea about that?. Scored correct: 1 dollar, [1].
- **How much of each \$100 of United States output was sold abroad?** (multiple choice). Choices: 1: 4 dollars worth, | 2: 8 dollars, | 3: 12 dollars, | 4: 24 dollars or more, or | 5: don't you have much idea about that?. Scored correct: 12 dollars, [3].
- **About how many adults per 100 in the most affected African countries had HIV/AIDS?** (multiple choice). Choices: 1: fewer than 5 out of 100, | 2: about 10 of 100, | 3: about 20 of 100, | 4: 30 or more out of 100, or | 5: don't you have much idea about that?. Scored correct: 30 or more out of 100, or [4].
- **Did the United States have a veto over World Trade Organization decisions?** (multiple choice). Choices: 1: true | 2: false. Scored correct: false [2].
- **Did the United States have a veto on the United Nations Security Council?** (multiple choice). Choices: 1: true | 2: false. Scored correct: true [1].
- **What was the main cause of global warming?** (multiple choice). Choices: 1: it is caused mostly by human activities, like driving cars a | 2: it is caused mostly by natural changes in the climate, | 3: it is not occurring at all, or | 4: haven't you thought much about this?. Scored correct: it is caused mostly by human activities, like driving cars a [1].

D.14 National Issues Convention 2 (2003)

- **Where did the Democratic Party stand on the global warming action scale?** (placement scale). Choices: 98: don't know. Scored correct: Above midpoint [6|7|8|9|10].
- **Where did the Republican Party stand on the global warming action scale?** (placement scale). Choices: 98: don't know. Scored correct: Below midpoint [0|1|2|3|4].

- **Did President Bush want to increase, decrease, or maintain foreign aid?** (multiple choice). Choices: 1: increase | 2: decrease | 3: keep it the same | 4: haven't heard anything about this | 8: don't know. Scored correct: increase [1].
- **Did President Bush support recent international agreements to control greenhouse gases?** (multiple choice). Choices: 1: supports | 3: opposes | 5: haven't heard anything about this | 8: don't know. Scored correct: opposes [3].
- **How much of each \$100 in the federal budget went to military spending?** (multiple choice). Choices: 1: 1 dollar | 2: 5 dollars | 3: 10 dollars | 4: 20 dollars | 5: 30+ dollars | 6: not much idea | 7: vol: other amount (specify) | 8: don't know. Scored correct: 20 dollars [4].
- **How much of each \$100 in the federal budget went to foreign aid?** (multiple choice). Choices: 1: 1 dollar | 2: 5 dollars | 3: 10 dollars | 4: 20 dollars | 5: 30+ dollars | 6: not much idea | 7: vol: other amount (specify) | 8: don't know. Scored correct: 1 dollar [1].
- **How much of each \$100 of United States output was sold abroad?** (multiple choice). Choices: 1: 4 dollars | 2: 8 dollars | 3: 12 dollars | 4: 24+ dollars | 5: not much idea | 7: vol: other amount (specify) | 8: don't know. Scored correct: 12 dollars [3].
- **About how many adults per 100 in the most affected African countries had HIV/AIDS?** (multiple choice). Choices: 1: fewer than 5 out of 100 | 2: about 10 of 100 | 3: about 20 of 100 | 4: 30+ out of 100 | 5: not much idea | 7: vol: other amount (specify) | 8: don't know. Scored correct: 30+ out of 100 [4].
- **Did the United States have a veto over World Trade Organization decisions?** (multiple choice). Choices: 1: true | 5: false | 8: don't know. Scored correct: false [5].
- **Did the United States have a veto on the United Nations Security Council?** (multiple choice). Choices: 1: true | 5: false | 8: don't know. Scored correct: true [1].
- **What was the main cause of global warming?** (multiple choice). Choices: 1: human activities | 2: natural changes | 3: not occurring at all | 4: haven't thought much about this | 8: don't know. Scored correct: human activities [1].

D.15 By the People 2004 Online Primaries (2004)

- **What did President Bush's tax cut do to taxes on investment income?** (multiple choice). Choices: 1: lower it | 2: keep it the same | 3: raise it | 4: don't know. Scored correct: lower it [1].
- **What was Wesley Clark's most recent government position?** (multiple choice). Choices: 1: senator from arkansas | 2: secretary of the air force | 3: supreme allied commander of nato | 4: national security advisor to the president | 5: don't know. Scored correct: supreme allied commander of nato [3].
- **Why did President Bush end tariffs on imported steel?** (multiple choice). Choices: 1: the steel industry and steelworkers unions objected | 2: he wanted to avoid increased tariffs by the european union o | 3: domestic steel production is insufficient to meet demand, th | 4: he has never supported tariffs | 5: don't know. Scored correct: he wanted to avoid increased tariffs by the european union o [2].

- **Which Democratic presidential candidate had served as House majority leader?** (multiple choice). Choices: 1: john edwards | 2: howard dean | 3: joe lieberman | 4: richard gephardt | 5: don't know. Scored correct: richard gephardt [4].
- **How had the unemployment rate changed since President Bush took office?** (multiple choice). Choices: 1: decreased | 2: stayed about the same | 3: increased | 4: don't know. Scored correct: increased [3].
- **How many Americans had been killed in action in Iraq?** (multiple choice). Choices: 1: fewer than a hundred | 2: several hundred | 3: several thousand | 4: more than twenty thousand | 5: don't know. Scored correct: several hundred [2].
- **What was the Bush administration's position on a Free Trade Area of the Americas?** (multiple choice). Choices: 1: support it | 2: neither support nor oppose it | 3: oppose it | 4: don't know. Scored correct: support it [1].

D.16 By the People 2004 US General Election (2004)

- **How did John Kerry vote on the Senate resolution authorizing the Iraq war?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: vote for it | 2: vote against it | 3: not vote on it | 4: blank | 5: don't know. Scored correct: vote for it [1].
- **Which country harbored the most Al Qaeda and Taliban fighters?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: india | 2: pakistan | 3: sri lanka | 4: indonesia | 5: blank | 6: don't know. Scored correct: pakistan [2].
- **Was Iraq directly involved in the September 11 attacks?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: true | 2: false | 3: blank | 4: don't know. Scored correct: false [2].
- **Had large quantities of weapons of mass destruction been found in Iraq?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: true | 2: false | 3: blank | 4: don't know. Scored correct: false [2].
- **Did prescription drugs cost more in Canada than in the United States?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: true | 2: false | 3: blank | 4: don't know. Scored correct: false [2].
- **What was George W. Bush's role during the Vietnam War?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: he was drafted but never went to vietnam | 2: he was a decorated officer serving in vietnam | 3: he was ineligible to serve in the military because of a medi | 4: he served in the texas air national guard | 5: blank | 6: don't know. Scored correct: he served in the texas air national guard [4].
- **What was John Kerry's role during the Vietnam War?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: he was drafted but never went to vietnam | 2: he was a decorated officer serving in vietnam | 3: he was ineligible to serve in the military because of a medi | 4: he served in the massachusetts air

national guard | 5: blank | 6: don't know. Scored correct: he was a decorated officer serving in vietnam [2].

- **Which country was a major destination for offshore white-collar jobs?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: south africa | 2: japan | 3: brazil | 4: india | 5: blank | 6: don't know. Scored correct: india [4].
- **About how many Americans had been killed in Iraq since the war began?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: 100 | 2: 500 | 3: 1,000 | 4: 10,000 | 5: blank | 6: don't know. Scored correct: 1,000 [3].

D.17 By the People: Health and Education (2005)

- **Where did the United States rank among wealthy countries in student mathematics skills?** (multiple choice). Choices: 1: bottom 3 | 2: bottom 10 | 3: top 10 | 4: top 3 | 7: dk. Scored correct: bottom 10 [2].
- **Had it become harder or easier for foreigners to get United States work permits?** (multiple choice). Choices: 1: harder | 2: same | 3: easier | 7: dk. Scored correct: harder [1].
- **Has the gap between minority students and white students in math and reading tests at the elementary school level been ...?** (multiple choice). Choices: 1: increasing the last few years | 2: staying about the same | 3: decreasing the last few years | 7: Couldn't say. Scored correct: decreasing the last few years [3]. Approved BTPHE-04: both questionnaires Q17 p.4 match education briefing PDF p.2; code 3 replaces inherited code 1 at both waves and calibration; original source responses unchanged.
- **About how many Americans lacked health insurance?** (multiple choice). Choices: 1: 1 million | 2: 25 million | 3: 45 million | 4: 65 million | 5: 85 million | 7: dk. Scored correct: 45 million [3].
- **How much was the population age 65 and older expected to grow over the next 50 years?** (multiple choice). Choices: 1: stay about the same | 2: increase by about half | 3: double | 4: triple | 7: dk. Scored correct: double [3].
- **What share of uninsured families had low incomes?** (multiple choice). Choices: 1: a quarter | 2: fifty per cent | 3: two thirds | 4: ninety per cent | 7: dk. Scored correct: two thirds [3].

D.18 Zeguo Township (2005)

- **By how much did Zeguo Township revenue increase from 2003 to 2004?** (multiple choice). Choices: 1: 10.2% | 2: 20.1% | 3: 33.7% | 4: No increase. Scored correct: 33.7% [3].
- **How large is the floating population in Zeguo Township?** (multiple choice). Choices: 1: 300,000 | 2: 50,000 | 3: 120,000 | 4: 200,000. Scored correct: 120,000 [3].
- **Which listed product is not a major product of Zeguo Township?** (multiple choice). Choices: 1: Water pumps | 2: Shoes | 3: Plastic products | 4: Air compressors. Scored correct: Plastic products [3].

- **How many parks does Zeguo Township currently have?** (multiple choice). Choices: 1: Zero | 2: One | 3: Two | 4: Five | 5: Seven. Scored correct: Two parks [3].

D.19 By the People 2007 (2007)

- **About how many Americans were prevented from voting because of criminal convictions?** (multiple choice). Choices: 1: 50,000 | 2: 500,000 | 3: 5,000,000 | 4: 15,000,000 | 99: Couldn't say. Scored correct: 5,000,000 [3].
- **What was the purpose of gerrymandering?** (multiple choice). Choices: 1: Increase electoral competition | 2: Ensure a majority for one party over another | 3: Include independents in the primary process | 4: Make them much smaller | 99: Couldn't say. Scored correct: Ensure a majority for one party over another [2].
- **What share of eligible voters voted in the specified presidential election?** (multiple choice). Choices: 1: 10% | 2: 30% | 3: 50% | 4: 70% | 99: Couldn't say. Scored correct: 50% [3].
- **Which listed country had compulsory voting?** (multiple choice). Choices: 1: Ireland | 2: United Kingdom | 3: Australia | 4: South Korea | 99: Couldn't say. Scored correct: Australia [3].
- **Who had to register with Selective Service in the United States?** (multiple choice). Choices: 1: Nobody | 2: Men between 18 and 25 years old | 3: Both men and women between 18 and 25 years old | 4: Men and women who receive federal grants for education | 99: Couldn't say. Scored correct: Men between 18 and 25 years old [2].
- **What vote total must a United States presidential candidate receive to win?** (multiple choice). Choices: 1: Gets more votes than any other candidate | 2: Gets a majority of all the votes cast | 3: Gets more Electoral College votes than any other candidate | 4: Gets a majority of Electoral College votes | 99: Couldn't say. Scored correct: Gets a majority of Electoral College votes [4].
- **How often are congressional districts redrawn?** (multiple choice). Choices: 1: Every 5 years | 2: Every 10 years | 3: Every 15 years | 4: Every 20 years | 99: Couldn't say. Scored correct: Every 10 years [2].
- **Which two states held the earliest presidential primary events?** (multiple choice). Choices: 1: Nevada and South Carolina | 2: Iowa and New Hampshire | 3: Indiana and New Mexico | 4: Wyoming and Delaware | 5: Michigan and Florida | 99: Couldn't say. Scored correct: Iowa and New Hampshire [2].

D.20 Northern Ireland (2007)

- **What percentage of majority-Protestant or majority-Catholic schools in Northern Ireland have at least 10% of the other religion in their enrolment?** (multiple choice). Choices: 1: more than 50% | 2: 40-50% | 3: 20-30% | 4: 5-10% | 5: less than 1% | 9: don't know. Scored correct: 5-10% [4]. NI-03: restored the omitted at-least-10%-other-religion condition. Baseline correct code 4 and departure correct code 3 both mean 5-10%; departure drops the baseline first option more than 50%. Existing keys and scores preserved. Remaining baseline choice text retains original dictionary transcription. Source

details: historical-cdd-scripts:legacy/poll_scripts/n_ireland.R lines 95/503; data/northern-ireland-2007/papers/northern-ireland-paper.pdf PDF p.8 Table 1; reports/northern-ireland-final-report.pdf PDF p.34

- **By about what percentage had the number of children entering Omagh schools changed?** (multiple choice). Choices: 1: increased by 20% | 2: increased by 10% | 3: stayed about the same | 4: decreased by 10% | 5: decreased by 20% | 9: don't know. Scored correct: decreased by 10% [4]. Archived variable/value labels are truncated; the listed substantive choices are partial transcriptions. The key is the archived numeric code.
- **What did the new entitlement framework require for the school curriculum?** (multiple choice). Choices: 1: every school provides all 14-year-olds with a choice of at 1 | 2: every child has the right to attend any school his or her pa | 3: every child be provided a free school meal every school day | 4: every child receives free textbooks | 9: don't know. Scored correct: every school provides all 14-year-olds with a choice of at 1... [1]. Archived variable/value labels are truncated; the listed substantive choices are partial transcriptions. The key is the archived numeric code.
- **What did the new entitlement framework require for access to courses?** (multiple choice). Choices: 1: every child receives tuition in the language of his or her p | 2: every child receives free transportation to and from school | 3: every denominational group has the right to run its own scho | 4: one-third of all the subjects offered must be applied | 9: don't know. Scored correct: one-third of all the subjects offered must be applied... [4]. Archived variable/value labels are truncated; the listed substantive choices are partial transcriptions. The key is the archived numeric code.
- **What did pupils in Northern Ireland typically do after compulsory schooling?** (multiple choice). Choices: 1: about one-quarter go directly into employment | 2: about one-quarter leave school to be unemployed | 3: about three-quarters of grammar school pupils attend univers | 4: about three-quarters of secondary school students attend fur | 9: don't know. Scored correct: about three-quarters of grammar school pupils attend univers... [3]. Archived variable/value labels are truncated; the listed substantive choices are partial transcriptions. The key is the archived numeric code.
- **Which statement about Northern Ireland school funding was correct?** (multiple choice). Choices: 1: schools receive more funding for older pupils | 2: schools receive the same funding for all pupils, regardless | 3: schools receive more funding for younger pupils | 9: don't know. Scored correct: schools receive more funding for older pupils [1]. Archived variable/value labels are truncated; the listed substantive choices are partial transcriptions. The key is the archived numeric code.
- **Which statement about the employing authority for Northern Ireland schools was correct?** (multiple choice). Choices: 1: the official employer for all teachers is the education and | 2: the official employer for all teachers is the department of | 3: the official employer for all teachers in catholic schools i | 4: the official employer for all teachers in voluntary grammar | 9: don't know. Scored correct: the official employer for all teachers in voluntary grammar... [4]. Archived variable/value labels are truncated; the listed substantive choices are partial transcriptions. The key is the archived numeric code.

D.21 Tomorrow's Europe (EU) (2007)

- **Which listed country was an official candidate to join the European Union?** (multiple choice). Choices: 1: Romania | 2: Montenegro | 3: Croatia | 4: Morocco | 5: Or couldn't you say about that ? | 6: Ref | 7: Dk. Scored correct: Croatia [3].
- **How were members of the European Parliament chosen?** (multiple choice). Choices: 1: Directly elected by the citizens of their country | 2: Elected for three year terms | 3: Elected by the parliament of their country | 4: Appointed by their national head of government | 5: Or couldn't you say about that ? | 6: Ref | 7: Dk. Scored correct: Directly elected by the citizens of their country [1].
- **Which European Union institutions adopted employment laws?** (multiple choice). Choices: 1: The European Commission, in some cases with the European Parliament | 2: The Council of Ministers, in some cases with the European Parliament | 3: The European Parliament by itself | 4: The European commission by itself | 5: Or couldn't you say about that ? | 6: Ref | 7: Dk. Scored correct: The Council of Ministers, in some cases with the European Parliament [2].
- **What was the European Union's role in unemployment benefits?** (multiple choice). Choices: 1: Finance the member states unemployment benefit systems | 2: Decide the level and length of unemployment benefits in the member states | 3: Require that the member states merge their unemployment benefit systems by 2010 | 4: Guarantee that all EU citizens have access to unemployment benefits where they live | 5: Or couldn't say about that ? | 6: Ref | 7: Dk. Scored correct: Guarantee that all EU citizens have access to unemployment benefits where they live [4].
- **To what was roughly one-third of the European Union budget devoted?** (multiple choice). Choices: 1: Helping the EU's less prosperous regions | 2: Subsidizing the EU's fishing industry | 3: Financing diplomatic missions abroad | 4: Maintaining the EU's administration and bureaucracy | 5: Or couldn't you say about that ? | 6: Ref | 7: Dk. Scored correct: Helping the EU's less prosperous regions [1].
- **How was the share of European Union adults age 65 or older expected to change by 2050?** (multiple choice). Choices: 1: About one quarter of what it is now | 2: About half of what it is now | 3: About the same as what it is now | 4: About twice what it is now | 5: About four times what it is now | 6: Or couldn't you say about that ? | 7: Ref | 8: Dk. Scored correct: About twice what it is now [4].
- **How much foreign aid did the European Union and its members give relative to the United States?** (multiple choice). Choices: 1: The EU and its member states give roughly four times as much | 2: The EU and its member states give roughly twice as much | 3: The EU and its member states and the US give about the same amount | 4: The US gives roughly twice as much | 5: The US gives roughly four times as much | 6: Or couldn't you say about that ? | 7: Ref | 8: Dk. Scored correct: The EU and its member states give roughly twice as much [2].
- **About what share of European Union member governments' spending went to foreign aid?** (multiple choice). Choices: 1: About 1% | 2: About 5% | 3: About 9% | 4: About 13% | 5: About 17% | 6: Or couldn't you say about that ? | 7: Ref | 8: Dk. Scored correct: About 1% [1].

- **Which listed country did not possess nuclear weapons?** (multiple choice). Choices: 1: Pakistan | 2: India | 3: North Korea | 4: Japan | 5: Or couldn't you say about that ? | 6: Ref | 7: Dk. Scored correct: Japan [4].
- **Where was Nicolas Sarkozy on the left to right political scale?** (placement scale). Choices: 1: 0 | 2: 1 | 3: 2 | 4: 3 | 5: 4 | 6: 5 | 7: 6 | 8: 7 | 9: 8 | 10: 9 | 11: 10 | 12: Ref | 13: Dk. Scored correct: 6 | 7 | 8 | 9 | 10 [7|8|9|10|11].
- **Where was Gordon Brown on the left to right political scale?** (placement scale). Choices: 1: 0 | 2: 1 | 3: 2 | 4: 3 | 5: 4 | 6: 5 | 7: 6 | 8: 7 | 9: 8 | 10: 9 | 11: 10 | 12: Ref | 13: Dk. Scored correct: 0 | 1 | 2 | 3 | 4 [1|2|3|4|5].

D.22 San Mateo, CA (2008)

- **What was the yearly gap between new homes needed and homes built in San Mateo County?** (multiple choice). Choices: 1: about 200 | 2: about 500 | 3: about 1,000 | 4: about 5,000 | 5: about 8,000 | 6: Or couldn't you say about that?. Scored correct: about 1,000 [3].
- **What was the median single-family home price in San Mateo County in 2007?** (multiple choice). Choices: 1: 550,000 | 2: 680,000 | 3: 750,000 | 4: 830,000 | 5: 950,000 | 6: Or couldn't you say about that?. Scored correct: 950,000 [5]. Source labels say \$950,000 for code 5; the March 12 pre-questionnaire says \$940,000. The administered version is not established; the numeric key remains 5.
- **What share of county households could afford a median-priced home in 2006?** (multiple choice). Choices: 1: 12% | 2: 23% | 3: 37% | 4: 43% | 5: 55% | 6: Or couldn't you say about that?. Scored correct: 12% [1].
- **What was the average rent for a two-bedroom apartment in 2007?** (multiple choice). Choices: 1: \$1,100 | 2: \$1,300 | 3: \$1,700 | 4: \$2,200 | 5: \$3,000 | 6: Or couldn't you say about that?. Scored correct: \$1,700 [3].
- **How much was county housing demand projected to grow from 2005 to 2025?** (multiple choice). Choices: 1: 20,000 | 2: 40,000 | 3: 60,000 | 4: 70,000 | 5: 90,000 | 6: Or couldn't you say about that?. Scored correct: 70,000 [4].
- **According to the same Housing Needs Study, among the new households that are expected to form in the County between 2005 and 2025, about what percentage, combined, would be households with low, very low and extremely low incomes?** (multiple choice). Choices: 1: Less than one-third | 2: Less than one-half | 3: About half | 4: More than one-half | 5: More than two-thirds | 6: Or couldn't you say about that?. Scored correct: About half [3]. SM-05 catalog wording repair: includes all three income groups named in Q24; correct code 3 and all scores unchanged. The retained onsite post questionnaire contains no knowledge battery, so departure wording is not independently established.
- **How many new homes were projected to be built at the current development rate?** (multiple choice). Choices: 1: 25,000 | 2: 50,000 | 3: 75,000 | 4: 100,000 | 5: Or couldn't you say about that?. Scored correct: 25,000 [1].
- **What percentage of San Mateo County's land is agricultural use, watershed, open space, wetlands, or parks?** (multiple choice). Choices: 1: less than 25% | 2: 33% | 3: 58%

| 4: 67% | 5: more than 75% | 6: Or couldn't you say about that?. Scored correct: more than 75% [5]. Question text is from the March 12 pre-questionnaire; answer codes and labels follow the source dictionary. The briefing chart groups agriculture/rangeland (40%) and open space (33%) separately and does not by itself establish the keyed threshold.

D.23 Europolis (2009)

- **V1Q43. Is the main decision-making body of the European Union the...?** (multiple choice). Choices: 1: European Commission | 2: Council of Ministers | 3: European Parliament | 4: European Court of Auditors | 999: You couldn't say. Scored correct: Council of Ministers [2].
- **V1Q44. Only one of the following statements about the European Parliament is false. Which one is it?** (multiple choice). Choices: 1: It passes all EU laws | 2: It can dismiss the European Commission | 3: It can reject the budget proposed by the Council of Minister | 4: It is involved in decisions about the admission of new Membe | 999: You couldn't say. Scored correct: It passes all EU laws [1].
- **V1Q46. Which of the following is true of Blue card workers?** (multiple choice). Choices: 1: They can work anywhere in the EU | 2: They must have a university education | 3: They cannot bring family members to join them any faster tha | 4: They are subject to the Returns Directive | 999: You couldn't say. Scored correct: They must have a university education [2].
- **V1Q47. Which of the following is true about the ways in which immigration policy is currently made?** (multiple choice). Choices: 1: The EU sets the basic rules about entry and residency requir | 2: The EU decides how many immigrants can be admitted to each c | 3: Work permits for immigrants must be approved by the EU | 4: The EU plays no role in immigration policy. | 999: You couldn't say. Scored correct: The EU sets the basic rules about entry and residency requir [1].
- **V1Q49. The percentage of the EU's total energy consumption that comes from fossil fuels (coal, gas or oil) is about...?** (multiple choice). Choices: 1: 50% | 2: 60% | 3: 70% | 4: 80% | 999: You couldn't say. Scored correct: 80% [4].
- **V1Q50. Which of the following produces the most greenhouse gases?** (multiple choice). Choices: 1: China | 2: The European Union | 3: The United States | 4: India | 999: You couldn't say. Scored correct: China [1].
- **Is the European Union represented on the international stage by the...?** (multiple choice). Choices: 1: European commission | 2: Council of ministers | 3: European Parliament | 4: European court of auditors. Scored correct: European Commission [1]. Arrival and departure only; expanded nine-item battery. Not observed at telephone baseline. Questions and correct answers corroborated by data/europolis-2009/reports/europolis-knowledge.pdf, p. 1, and data/europolis-2009/codebook.txt. Answer choices reproduce retained source value labels; some are abbreviated, and their missing wording is not reconstructed. The former Europolis questionnaire-post.pdf was a misfiled content duplicate of the Tomorrow's Europe 2007 post questionnaire and was removed; the canonical form remains under data/tomorrows-europe-2007/. It does not document these Europolis questions.
- **Which of the following is true of the EU's immigrants?** (multiple choice). Choices:

1: Most illegal immigrants enter the EU legally | 2: Roughly 9% of EU's total population | 3: Because of immigration the EU total | 4: There are more illegal than legal. Scored correct: Most illegal immigrants enter the EU legally but outstay their visas [1]. Arrival and departure only; expanded nine-item battery. Not observed at telephone baseline. Questions and correct answers corroborated by data/europolis-2009/reports/europolis-knowledge.pdf, p. 1, and data/europolis-2009/codebook.txt. Answer choices reproduce retained source value labels; some are abbreviated, and their missing wording is not reconstructed. The former Europolis questionnaire-post.pdf was a misfiled content duplicate of the Tomorrow's Europe 2007 post questionnaire and was removed; the canonical form remains under data/tomorrows-europe-2007/. It does not document these Europolis questions.

- **Which of the following is true about wind power in the European Union?** (multiple choice). Choices: 1: Produced off-shore | 2: Increasing 30% | 3: Three times of solar power | 4: Consumption about 5%. Scored correct: Its share of energy consumption is increasing about 30 percent a year [2]. Arrival and departure only; expanded nine-item battery. Not observed at telephone baseline. Questions and correct answers corroborated by data/europolis-2009/reports/europolis-knowledge.pdf, p. 1, and data/europolis-2009/codebook.txt. Answer choices reproduce retained source value labels; some are abbreviated, and their missing wording is not reconstructed. The former Europolis questionnaire-post.pdf was a misfiled content duplicate of the Tomorrow's Europe 2007 post questionnaire and was removed; the canonical form remains under data/tomorrows-europe-2007/. It does not document these Europolis questions.

D.24 Michigan (2009)

- **Which party held the majority in the Michigan State Senate?** (open text). Scored correct: Republican Party [2]. Open party name at departure was lowercased and stripped of punctuation; party-name variants listed in the cited dataset were accepted. The baseline source stores a coded party category, shown in the answer key.
- **Which party held the majority in the Michigan State House of Representatives?** (open text). Scored correct: Democratic Party [1]. Open party name at departure was lowercased and stripped of punctuation; party-name variants listed in the cited dataset were accepted. The baseline source stores a coded party category, shown in the answer key.
- **Which state had an unemployment rate similar to Michigan?** (multiple choice). Choices: 1: Oregon | 2: West Virginia | 3: Ohio | 4: Louisiana. Scored correct: Oregon [1].
- **Under which condition could a person remain eligible for the Family Independence Program after 48 months?** (multiple choice). Choices: 1: Live in a county with high unemployment | 2: Fired from a job while on the Family Self-Sufficiency Plan | 3: Non-citizen on the Refugee Assistance Program | 4: Have a small child in the ChildFirst program. Scored correct: Live in a county with high unemployment [1].
- **About what percentage of African American children in Michigan lived in poverty?** (multiple choice). Choices: 1: Under 20% | 2: About 30% | 3: About 40% | 4: Over 50%. Scored correct: About 40% [3].
- **Where was the Democratic Party on the 1–7 scale from lower taxes and services to higher taxes and services?** (placement scale). Choices: 1–7 placement scale; endpoints in question. Scored correct: Positions 5, 6, 7 [5|6|7].

- **Where was the Republican Party on the 1–7 scale from lower taxes and services to higher taxes and services?** (placement scale). Choices: 1–7 placement scale; endpoints in question. Scored correct: Positions 1, 2, 3 [1|2|3].
- **Where was the Democratic Party on the 1–7 scale from government action to reduce unemployment to letting the economy recover?** (placement scale). Choices: 1–7 placement scale; endpoints in question. Scored correct: Positions 1, 2, 3 [1|2|3].
- **Where was the Republican Party on the 1–7 scale from government action to reduce unemployment to letting the economy recover?** (placement scale). Choices: 1–7 placement scale; endpoints in question. Scored correct: Positions 5, 6, 7 [5|6|7].
- **Where would you place the Democratic Party on the same scale?** (party placement). Choices: 1: See to it that everyone has a job and certain standard of living | 2 | 3 | 4: Exactly in the middle | 5 | 6 | 7: Let everyone try to get ahead on his or her own | 99: No opinion. Scored correct: Government side [1|2|3]. Arrival and departure only; six-placement supplemental battery. Never substitute arrival for telephone baseline. Questionnaire p. 3, Q7–8 use the State Government responsibility scale introduced in Q6. Correctness uses Democratic placement on the government side (1–3) and Republican placement on the individual side (5–7); the midpoint and no opinion are incorrect. Arrival counts 196/310 and 209/310 reproduce the percentages in data/michigan-2009/reports/michigan-final-report.pdf, p. 13.
- **Where would you place the Republican Party on the same scale?** (party placement). Choices: 1: See to it that everyone has a job and certain standard of living | 2 | 3 | 4: Exactly in the middle | 5 | 6 | 7: Let everyone try to get ahead on his or her own | 99: No opinion. Scored correct: Individual side [5|6|7]. Arrival and departure only; six-placement supplemental battery. Never substitute arrival for telephone baseline. Questionnaire p. 3, Q7–8 use the State Government responsibility scale introduced in Q6. Correctness uses Democratic placement on the government side (1–3) and Republican placement on the individual side (5–7); the midpoint and no opinion are incorrect. Arrival counts 196/310 and 209/310 reproduce the percentages in data/michigan-2009/reports/michigan-final-report.pdf, p. 13.

D.25 California What’s Next (2011)

- **Which party held the majority in the California State Senate?** (multiple choice). Choices: 1: Republican Party | 2: Democratic Party | 98: Don’t Know/ Couldn’t Say. Scored correct: Democratic Party [2].
- **Which party held the majority in the California State Assembly?** (multiple choice). Choices: 1: Republican Party | 2: Democratic Party | 98: Don’t Know/ Couldn’t Say. Scored correct: Democratic Party [2].
- **What legislative majority was needed to approve the proposed state budget?** (multiple choice). Choices: 1: Simple Majority of Both Houses | 2: Simple Majority Assembly & Two-Thirds Senate | 3: Two-Thirds of Both Houses | 4: Three-Fourths of Both Houses | 98: Don’t Know/ Couldn’t Say. Scored correct: Two-Thirds of Both Houses [3].
- **What legislative majority was needed to raise taxes?** (multiple choice). Choices: 1: Simple Majority of Both Houses | 2: Simple Majority Assembly & Two-Thirds Senate | 3: Two-Thirds of Both Houses | 4: Three-Fourths of Both Houses | 98: Don’t Know/ Couldn’t Say. Scored correct: Two-Thirds of Both Houses [3].

- **Who could sign a ballot measure?** (multiple choice). Choices: 1: Anyone 18+ | 2: Any Citizen 18+ | 3: Only Registered Voters | 4: Only Registered Voters Who Voted in Last Election | 98: Don't Know/ Couldn't Say. Scored correct: Only Registered Voters [3].
- **Which state has the most residents per member of the state legislature?** (multiple choice). Choices: 1: California | 2: Texas | 3: New York | 4: Florida | 5: Couldn't say. Scored correct: California [1]. Arrival and exit expanded eight-item battery only; not asked in this form at telephone baseline. Questionnaire pp. 17–18, Q32–34; keys corroborated by data/california-whats-next-2011/reports/california-report.pdf, Table 4.
- **On average, which state has the highest total tax burden?** (multiple choice). Choices: 1: California | 2: New York | 3: Massachusetts | 4: Oregon | 5: Couldn't say. Scored correct: New York [2]. Arrival and exit expanded eight-item battery only; not asked in this form at telephone baseline. Questionnaire pp. 17–18, Q32–34; keys corroborated by data/california-whats-next-2011/reports/california-report.pdf, Table 4.
- **In Governor Brown's most recent budget proposal, the largest single share of spending goes to which of the following?** (multiple choice). Choices: 1: Higher Education | 2: K-12 education | 3: Health and Human Services | 4: State Pensions | 5: Couldn't say. Scored correct: K-12 education [2]. Arrival and exit expanded eight-item battery only; not asked in this form at telephone baseline. Questionnaire pp. 17–18, Q32–34; keys corroborated by data/california-whats-next-2011/reports/california-report.pdf, Table 4.

D.26 America in One Room (2019)

- **Which political party holds the majority in the Senate?** (multiple choice). Choices: 1: Democratic | 2: Republican | 3: Independent. Scored correct: Republican [2].
- **Which political party holds the majority in the House?** (multiple choice). Choices: 1: Democratic | 2: Republican | 3: Independent. Scored correct: Democratic [1].
- **About how many undocumented immigrants are in the US?** (multiple choice). Choices: 1: 10 million | 2: 20 million | 3: 30 million | 4: 40 million. Scored correct: 10 million [1].
- **Which of the following countries is NOT part of the Paris Agreement on the environment?** (multiple choice). Choices: 1: Russia | 2: Turkey | 3: Iran | 4: All of the above. Scored correct: All of the above [4].
- **The Affordable Care Act allows which of the following?** (multiple choice). Choices: 1: Stay on your parents' healthcare plan until age 26 | 2: Eliminated annual and lifetime limits on insurance coverage | 3: Cannot charge women higher rates than men | 4: All of the above. Scored correct: All of the above [4].
- **What percentage is the highest tax rate for capital gains taxes?** (multiple choice). Choices: 1: Under 10% | 2: Around 20% | 3: Around 30% | 4: More than 40%. Scored correct: Around 20% [2].
- **Which of the following organizations dealing with trade has the most countries?** (multiple choice). Choices: 1: North American Free Trade Agreement (NAFTA) | 2: World Trade Organization (WTO) | 3: United States-Mexico-Canada Agreement (USMCA) | 4: Trans-Pacific Partnership (TPP). Scored correct: World Trade Organization (WTO) [2].

D.27 America in One Room: Climate (2021)

- **Which political party controls the majority in the Senate?** (multiple choice). Scored correct: Code 1 (answer text not recovered). Offered choice labels were not in the retained report; the keyed code is checked against reported item results.
- **Which political party controls the majority in the House?** (multiple choice). Scored correct: Code 1 (answer text not recovered). Offered choice labels were not in the retained report; the keyed code is checked against reported item results.
- **Which of the following best describes Net Zero?** (multiple choice). Choices: 1: Capturing fossil fuels emissions with a special net in the atmosphere | 2: Storing emissions underground | 3: Not adding to the total amount of climate-heating gases in the atmosphere | 4: Using the same amount of renewable energy and fossil fuels so we don't use too much from one source of energy | 77: Couldn't say. Scored correct: Not adding to the total amount of climate-heating gases in the atmosphere [3]. Question options and keys recovered from original authored preparation script, lines 355-424. Code 77 means couldn't say; administrative codes 98 (skipped on web) and 99 (refused) are not offered answer choices. All three are non-substantive; preserve their raw codes and score zero in observed questionnaires. Existing keys and scores unchanged. Source script uses six climate-fact items; the canonical eight-item battery also retains the separately documented party-control questions Q17/Q18.
- **What is the name of a method by which government can encourage emitters to reduce greenhouse gases?** (multiple choice). Choices: 1: Carbon pricing | 2: Carbon capture | 3: Carbon sequestration | 4: Carbon coin | 77: Couldn't say. Scored correct: Carbon pricing [1]. Question options and keys recovered from original authored preparation script, lines 355-424. Code 77 means couldn't say; administrative codes 98 (skipped on web) and 99 (refused) are not offered answer choices. All three are non-substantive; preserve their raw codes and score zero in observed questionnaires. Existing keys and scores unchanged. Source script uses six climate-fact items; the canonical eight-item battery also retains the separately documented party-control questions Q17/Q18.
- **Which of the following emit greenhouse gas emissions?** (multiple choice). Choices: 1: Methane | 2: Coal | 3: Oil | 4: Natural gas | 5: All of the above | 77: Couldn't say. Scored correct: All of the above [5]. Question options and keys recovered from original authored preparation script, lines 355-424. Code 77 means couldn't say; administrative codes 98 (skipped on web) and 99 (refused) are not offered answer choices. All three are non-substantive; preserve their raw codes and score zero in observed questionnaires. Existing keys and scores unchanged. Source script uses six climate-fact items; the canonical eight-item battery also retains the separately documented party-control questions Q17/Q18.
- **Which country emits the most greenhouse gas emissions?** (multiple choice). Choices: 1: China | 2: Us | 3: Japan | 4: Brazil | 77: Couldn't say. Scored correct: China [1]. Question options and keys recovered from original authored preparation script, lines 355-424. Code 77 means couldn't say; administrative codes 98 (skipped on web) and 99 (refused) are not offered answer choices. All three are non-substantive; preserve their raw codes and score zero in observed questionnaires. Existing keys and scores unchanged. Source script uses six climate-fact items; the canonical eight-item battery also retains the separately documented party-control questions Q17/Q18.
- **The term fossil fuels refers to which of the following energy sources?** (multiple

choice). Choices: 1: Hydrocarbons | 2: Biofuels | 3: New generation nuclear | 4: Hydrogen | 77: Couldn't say. Scored correct: Hydrocarbons [1]. Question options and keys recovered from original authored preparation script, lines 355-424. Code 77 means couldn't say; administrative codes 98 (skipped on web) and 99 (refused) are not offered answer choices. All three are non-substantive; preserve their raw codes and score zero in observed questionnaires. Existing keys and scores unchanged. Source script uses six climate-fact items; the canonical eight-item battery also retains the separately documented party-control questions Q17/Q18.

- **Which international agreement is about climate change?** (multiple choice). Choices: 1: Paris Agreement | 2: Trans-Pacific Partnership | 3: United Nations Relief Partnership | 4: G20 Agreement on Pillars One and Two | 77: Couldn't say. Scored correct: Paris Agreement [1]. Question options and keys recovered from original authored preparation script, lines 355-424. Code 77 means couldn't say; administrative codes 98 (skipped on web) and 99 (refused) are not offered answer choices. All three are non-substantive; preserve their raw codes and score zero in observed questionnaires. Existing keys and scores unchanged. Source script uses six climate-fact items; the canonical eight-item battery also retains the separately documented party-control questions Q17/Q18.